



كيف تكتب ورقة مراجعة منهجية واحترافية وتسرع قبول نشرك العلمي

How to Write a Professional Systematic Literature Review (SLR) and Speed Up Acceptance of Your Scientific Publication

Presented by

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He holds a **PhD** in Information Technology from UTP (Malaysia), an **MSc** in Computer Applications from Bangalore University (India), and a **BEng** in Software Engineering from Taiz University (Yemen).

With over ten years of combined experience in academia and industry, his research interests span Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Metaheuristic Algorithms, and Optimization, with additional expertise in Android and web development.

[LinkedIn](#) | [Google Scholar](#) | [Scopus](#) | [ResearchGate](#) | [Web of Science](#) | [ORCID](#) | [GitHub](#) | [Tutorials](#)

Prerequisites

Introduction

- Literature vs Literature Review
- What, Why, and How do we conduct SLRs?
- Survey, literature, systematic literature review, and meta-analysis.

Methodology

The literature study mapping process

Step 1: Preliminary study

Step 2: Screening process

Step 3: Eligibility and quality assessment

Step 4: Data extraction

Extra

Achievements

Don't reinvent the wheel...

Implication of incomplete literature review!



OPTIMIZED LONG SHORT-TERM MEMORY WITH AN ADAPTIVE AQUILA OPTIMIZER FOR LONG-TERM DEPENDENCY PROBLEMS

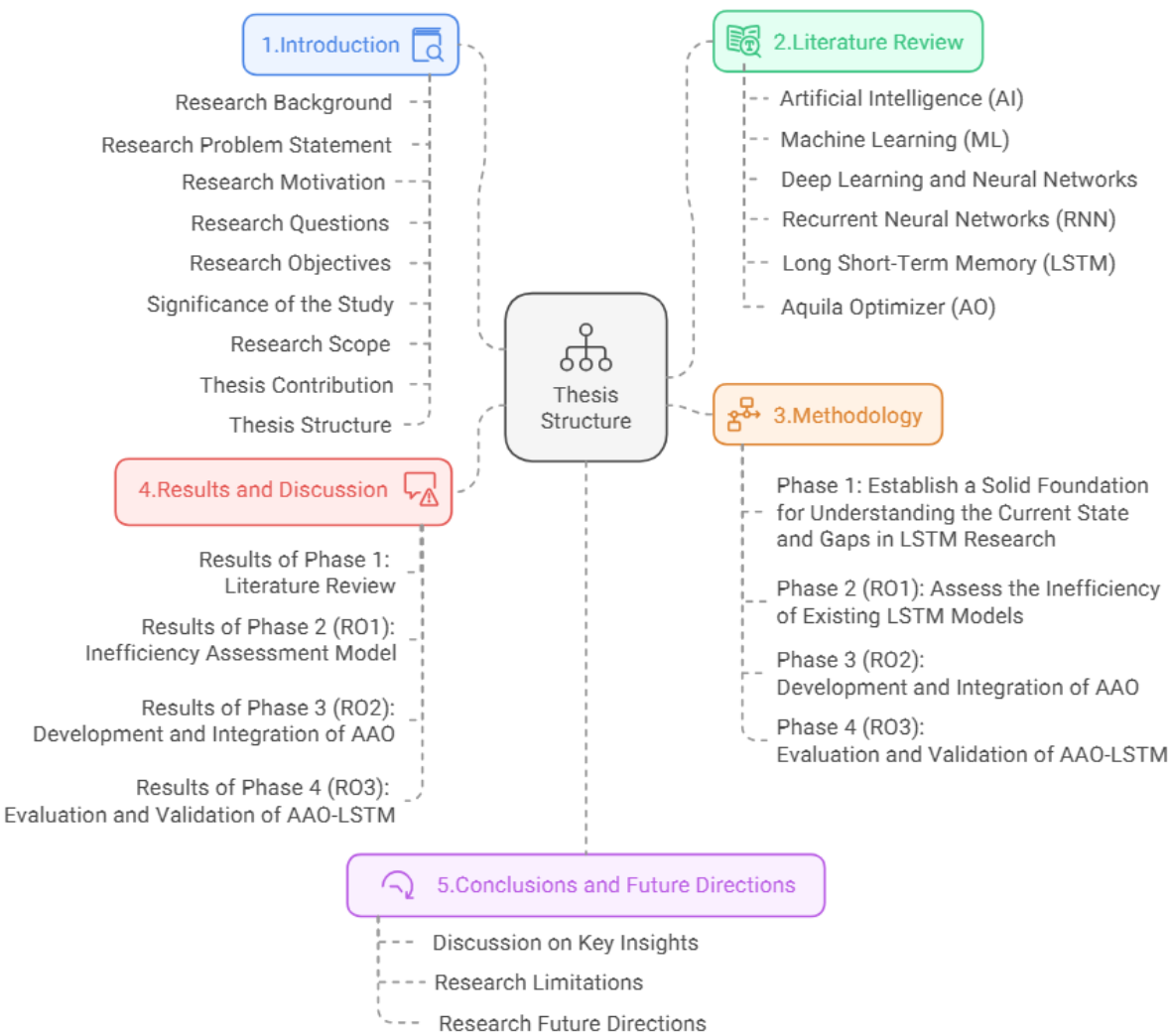


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1. *Al-Selwi, S. M., Hassan, M. F., Abdulkadir, S. J., Muneer, A., Sumiea, E. H., Alqushaibi, A., & Ragab, M. G. (2024). *RNN-LSTM: From Applications to Modeling Techniques and Beyond - Systematic Review*. *Journal of King Saud University-Computer and Information Sciences*, 102068. (ISI-indexed **Q1** Journal, Impact Factor 6.1)
2. *Sumiea, E. H., Abdulkadir, S. J., Alhussian, H. S., Al-Selwi, S. M., Alqushaibi, A., Ragab, M. G., & Fati, S. M. (2024). *Deep Deterministic Policy Gradient Algorithm: A Systematic Review*. *Heliyon*. (ISI-indexed **Q1** Journal, Impact Factor 3.6)
3. *Ragab, M. G., Abdulkader, S. J., Muneer, A., Alqushaibi, A., Sumiea, E. H., Qureshi, R., Al-Selwi, S. M., & Alhussian, H. (2024). *A Comprehensive Systematic Review of YOLO for Medical Object Detection (2018 to 2023)*. *IEEE Access*. (ISI-indexed **Q2** Journal, Impact Factor 3.6)
4. *Hassan, S. U., Abdulkadir, S. J., Zahid, M. S. M., & Al-Selwi, S. M. (2025). *Local Interpretable Model-Agnostic Explanation Approach for Medical Imaging Analysis: A Systematic Literature Review*. *Computers in Biology and Medicine*, 185, 109569. (ISI-indexed **Q1** Journal, Impact Factor 6.3)
5. *Yalli, J. S., Hasan, M. H., Jung, L. T., & Al-Selwi, S. M. (2025). *Authentication Schemes for Internet of Things (IoT) Networks: A Systematic Review and Security Assessment*. *Internet of Things*, 101469. (ISI-indexed **Q1** Journal, Impact Factor 7.6)
6. *Yalli, J. S., Hasan, M. H., Jung, L. T., Yerima, A. I., Aliyu, D. A., Maiwada, U. D., Al-Selwi, S. M., & Shaikh, M. U. R. (2025). *A systematic review for evaluating IoT security: A focus on authentication, protocols, and enabling technologies*. *IEEE Internet of Things Journal*. (ISI-indexed **Q1** Journal, Impact Factor 8.2)
7. *Fayyaz, A., Abdulkadir, S. J., Hassan, S. U., Al-Selwi, S. M., Sumiea, E. H. H., & Talib L. F. (2025). *The Role of Advanced Machine Learning in COVID-19 Medical Imaging: A Technical Review*. *Results in Engineering*, 105154. (ISI-indexed **Q1** Journal, Impact Factor 7.9).
8. Fayyaz, A. M., Abdulkadir, S. J., Talpur, N., Al-Selwi, S. M., Hassan, S. U., & Sumiea, E. H. (2025). *Grad-CAM (Gradient-weighted Class Activation Mapping): A systematic literature review*. *Computers in Biology and Medicine*, 198, 111200. (ISI-indexed **Q1** Journal, Impact Factor 6.3)
9. *Vikash, M. et al. *Machine Learning in Credit Card Fraud Detection: A Systematic Review with Emphasis on Concept Drift*. (Under Review).
10. *Ayaz et al. *Grain Disease Detection and Classification using Advanced Computer Vision Techniques: A Technical Review*. (Under Review)

Data	Microsoft Office - Excel
Reference manager	EndNote , Zotero , Mendeley , Rayyan AI
Bibliometric Mapping	VOSviewer
Code	https://github.com/SafwanAlselwi/RNN-LSTM-SLR
SLR paper to follow	Al-Selwi, S. M., Hassan, M. F., Abdulkadir, S. J., & et al. (2024). RNN-LSTM: From applications to modeling techniques and beyond—Systematic review . <i>Journal of King Saud University-Computer and Information Sciences</i> , 36(5), 102068. https://doi.org/10.1016/j.jksuci.2024.102068

Introduction



Literature vs. Literature Review

• A Simple Comparison for Researchers •



LITERATURE



What is it?

All published sources and information on a topic.



What does it include?

Books, articles, reports, conference papers, theses, and more.



Purpose

Provides information and builds background knowledge.



Key Question

What is already known?



LITERATURE REVIEW



What is it?

A systematic process of finding, evaluating, and synthesizing studies.



What does it include?

Relevant studies selected, analyzed, and compared.



Purpose

Summarizes evidence, identifies patterns, gaps, and draws conclusions.



Key Question

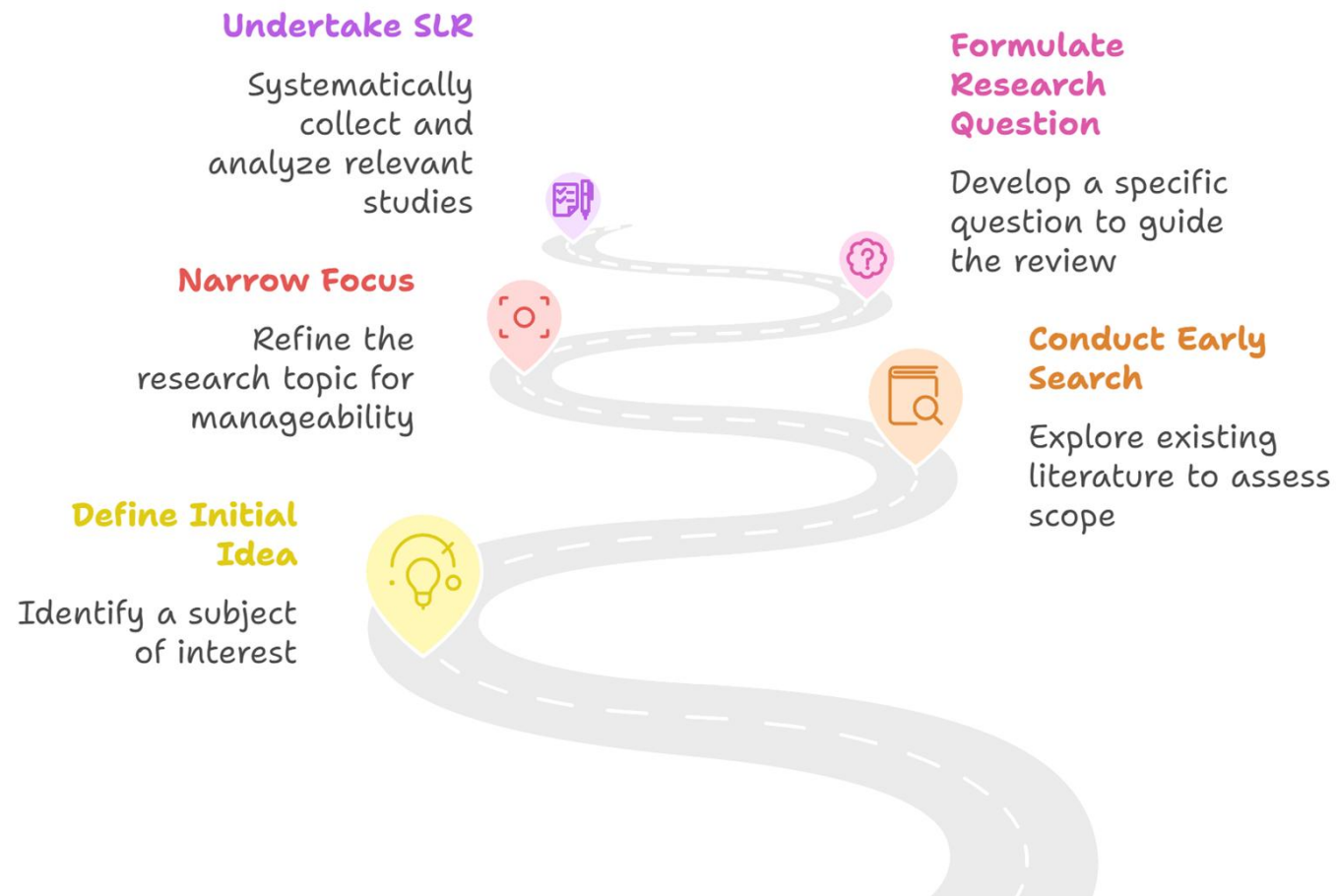
What does the evidence show and what are the gaps?

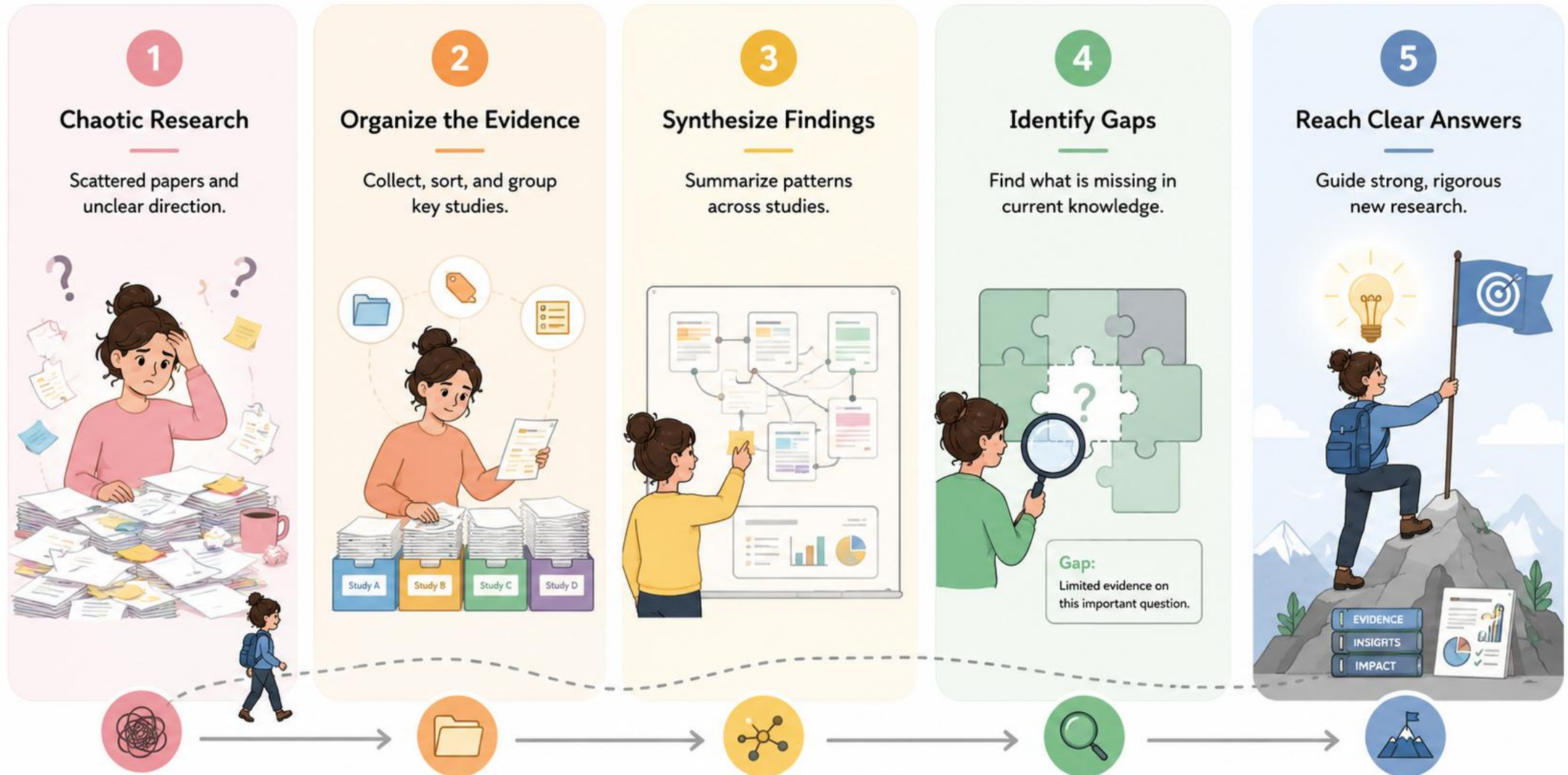


In Summary: Literature is everything available on a topic.
Literature review is the process of organizing and analyzing it to generate new insights.

An **SLR** is a research methodology to identify, collect, and critically analyse the available research studies through a systematic procedure to *answer a set of pre-defined research questions*.

Systematic Literature Review Process



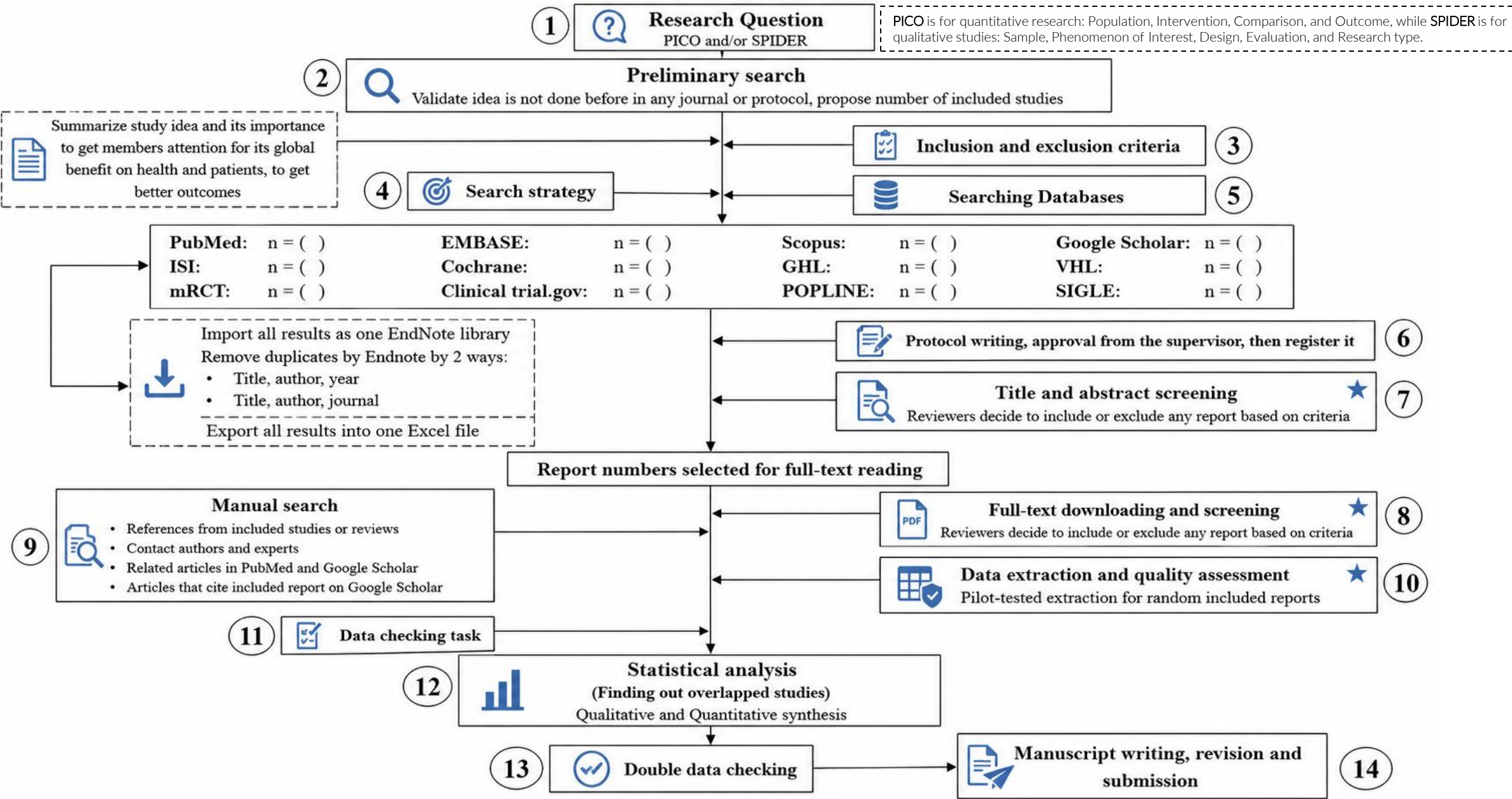


Goal: To **identify gaps** in the literature and **support these gaps with numerical data**.

A simple comparison of Survey, Traditional Literature Review, SLR, and Meta-analysis

	 Survey	 Traditional Literature Review	 Systematic Literature Review (SLR)	 Meta-analysis
 Purpose	Broad overview of a research area, methods, or trends	Summarize and discuss prior studies related to a topic	Answer specific research questions using a structured process	Statistically combine findings from comparable quantitative studies
 Search approach	Selective and flexible; not always fully documented	Narrative and non-systematic; sources chosen by the author	Comprehensive, protocol-based, with explicit inclusion/exclusion criteria	Usually follows a systematic review process first
 Analysis style	Mostly descriptive and thematic	Critical discussion and interpretation	Structured synthesis of selected studies	Quantitative synthesis using effect sizes and statistics
 Strengths	Easy to read; useful for understanding the landscape	Builds background and context for a study	Transparent, rigorous, and reproducible	Produces stronger overall evidence and pooled estimates
 Limitations	May reflect author bias; less reproducible	Search may be incomplete; difficult to replicate	Time-consuming; requires careful planning	Needs sufficient comparable data; heterogeneity can be a challenge
 Quick takeaway	 Broad descriptive overview	 Narrative interpretation of prior work	 Systematic and reproducible evidence synthesis	 Statistical synthesis of quantitative evidence

Methodology





PRISMA

Preferred Reporting Items for Systematic Reviews and Meta-Analyses

A reporting guideline that helps researchers report systematic reviews and meta-analyses in a transparent, complete, and reliable way.



1. What is PRISMA?

- An evidence-based reporting guideline
- Used to improve clarity and transparency in reviews
- Supports complete and structured reporting



2. Why is it important?

- Increases transparency
- Improves reproducibility
- Helps readers assess review quality
- Builds trust in the evidence



3. What does PRISMA include?

- PRISMA Flow Diagram
- PRISMA Checklist
- Guidance for reporting each review stage



4. Where is it used?

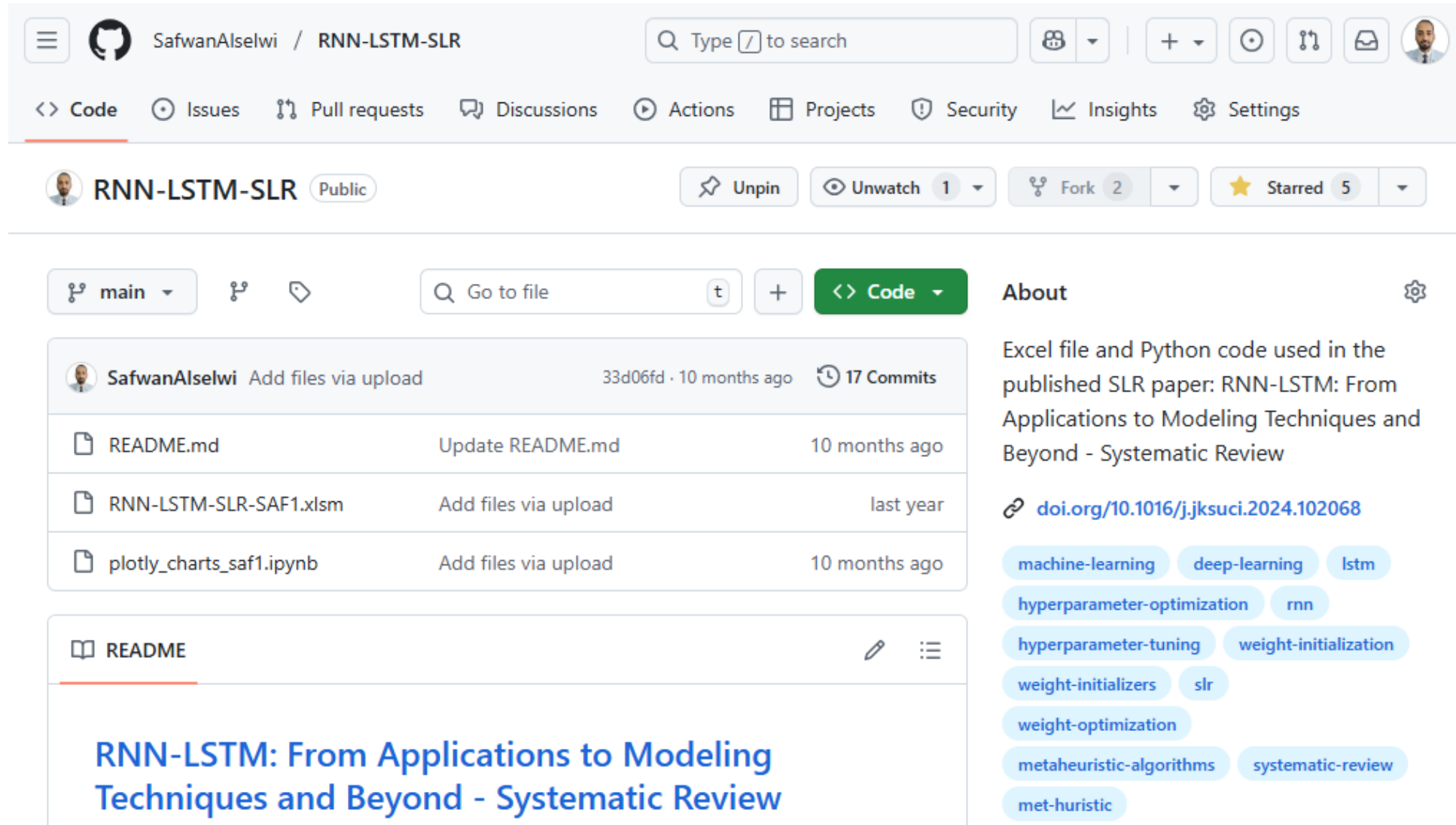
- Systematic Literature Reviews (SLR)
- Systematic Reviews
- Meta-Analyses
- Scoping Reviews



Key Takeaway

PRISMA helps researchers report reviews clearly, consistently, and transparently.

<https://github.com/SafwanAlselwi/RNN-LSTM-SLR>



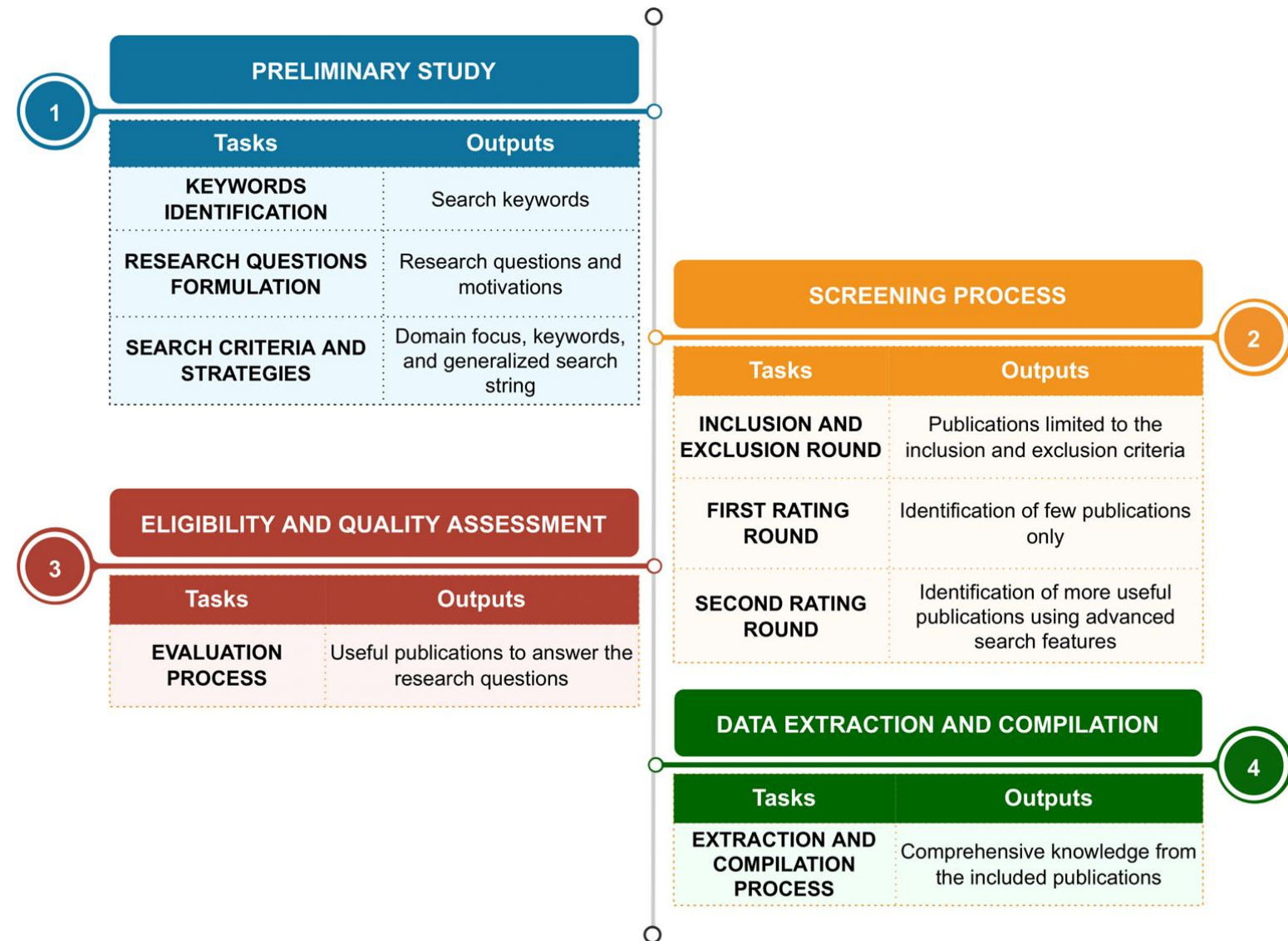
The screenshot shows the GitHub repository page for 'RNN-LSTM-SLR' by user 'SafwanAlselwi'. The repository is public and has 5 stars, 2 forks, and 1 unwatch. The main branch is 'main'. The repository contains three files: 'README.md' (updated 10 months ago), 'RNN-LSTM-SLR-SAF1.xlsm' (added last year), and 'plotly_charts_saf1.ipynb' (added 10 months ago). The README is titled 'RNN-LSTM: From Applications to Modeling Techniques and Beyond - Systematic Review'. The repository description states: 'Excel file and Python code used in the published SLR paper: RNN-LSTM: From Applications to Modeling Techniques and Beyond - Systematic Review'. The DOI is 'doi.org/10.1016/j.jksuci.2024.102068'. The repository is tagged with various keywords: machine-learning, deep-learning, lstm, hyperparameter-optimization, rnn, hyperparameter-tuning, weight-initialization, weight-initializers, slr, weight-optimization, metaheuristic-algorithms, systematic-review, and met-huristic.

The SLR was carried out following the guidelines provided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) (Page et al., 2021).

The study also adhered to the standard recommendations proposed by Kitchenham and Charters (2007).

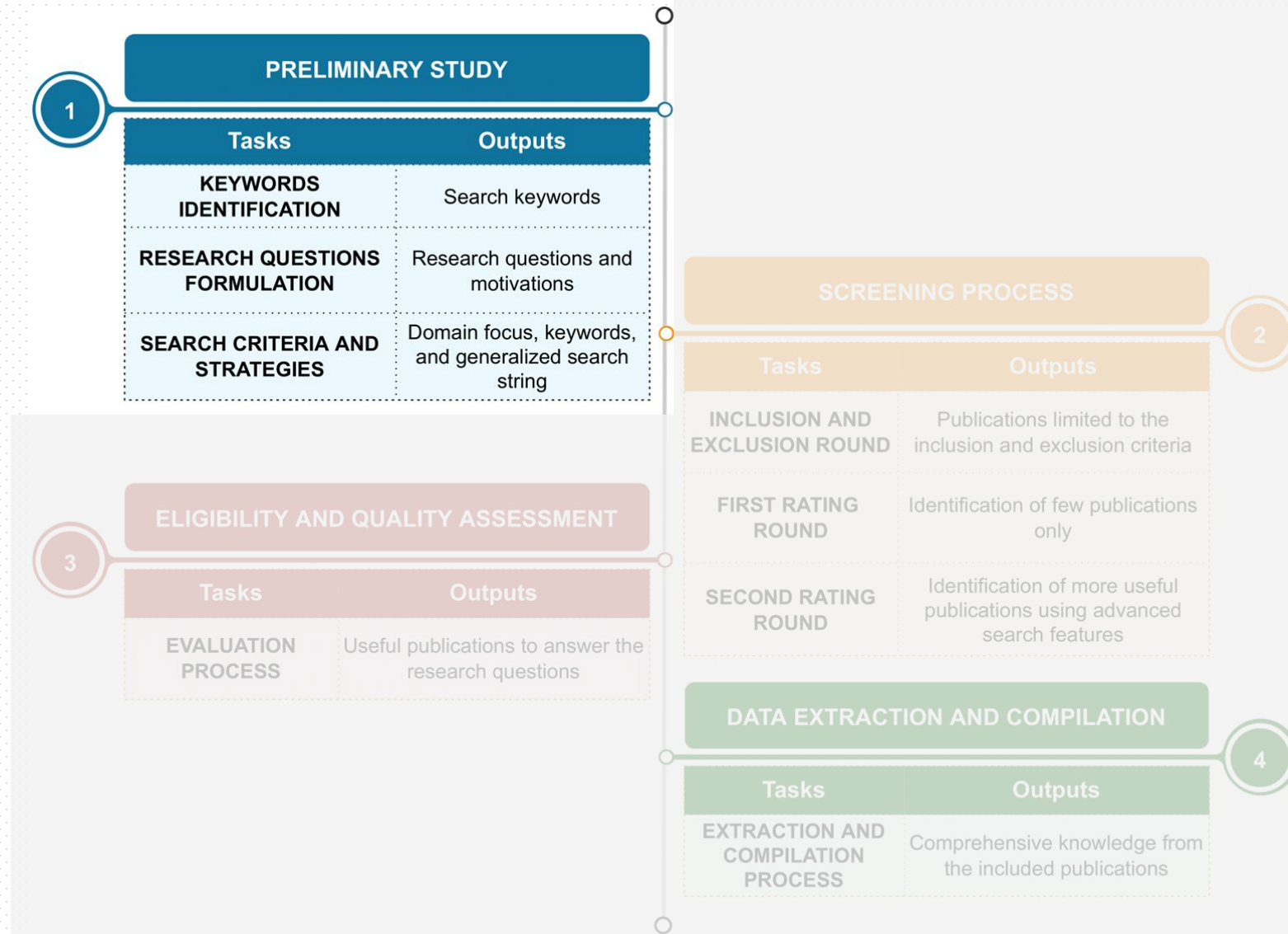
Additionally, to ensure comprehensive reporting, the structure, and approach of previous extensive studies were adopted (Abdullah et al., 2023, Talpur et al., 2023)

- [PRISMA statement](#)
- [PRISMA2020 flow diagrams](#)



Step 1

Preliminary study



Step 1: The Preliminary study

An initial exploration is conducted to build a deeper understanding of the main topic. This step helps researchers define strong research questions, identify relevant keywords, and clarify the scope of the review.

Tasks	Outputs
1 Keywords Identification	Search keywords and related terms
2 Research Questions Formulation	Clear SLR research questions and motivations
3 Search Criteria and Strategies	Domain focus, scope, keywords, and generalized search string



Purpose: To establish a strong foundation for the review before screening begins.



Step 1.1 Keywords identification & Boolean operators

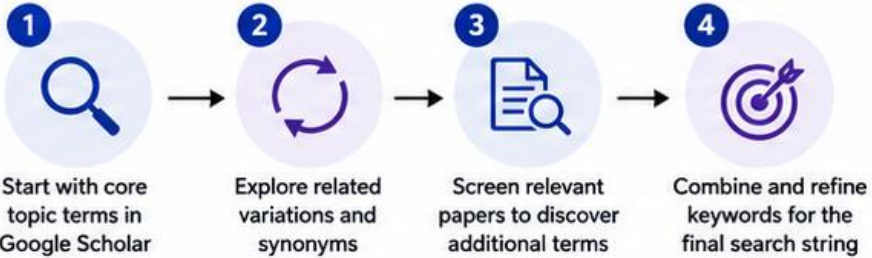


How to refine search terms for a systematic literature review



Keywords identification is the process of selecting, testing, and refining search terms that best capture the topic of interest and retrieve the most relevant studies.

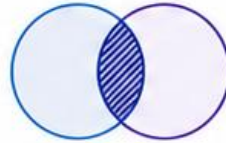
PROCESS: HOW TO IDENTIFY AND REFINED KEYWORDS



BOOLEAN OPERATORS

AND

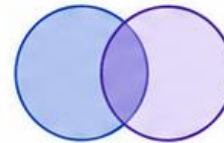
Narrows the search by combining concepts



Example:
LSTM AND weight

OR

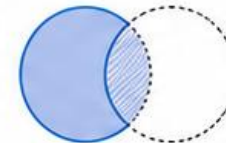
Broadens the search by including synonyms or alternatives



Example:
LSTM OR
"long short term memory"

NOT

Excludes unwanted terms



Example:
LSTM NOT survey

" "

Search for an exact phrase



Example:
"long short term memory"

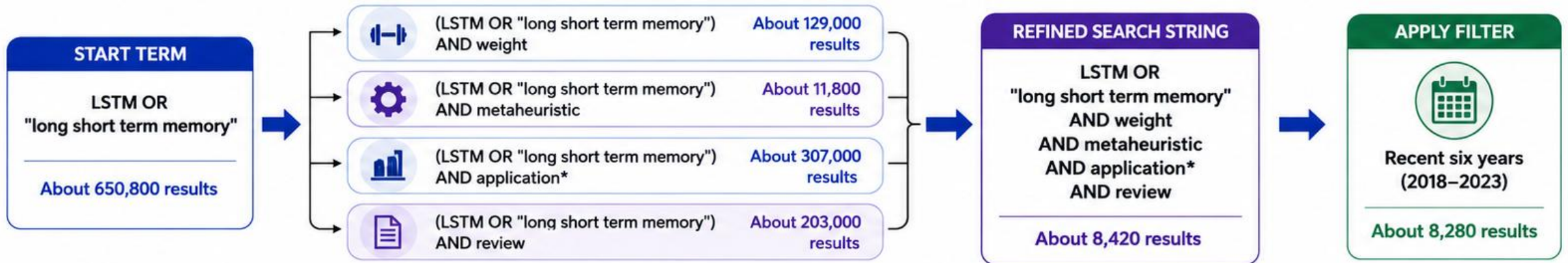
*

Captures word variations



Example:
optim* → optimization,
optimizer, optimizing

SEARCH REFINEMENT EXAMPLE: LSTM CASE



TAKEAWAY: Well-chosen keywords improve search accuracy, reduce irrelevant results, and help define the scope of the review. [Scopus Support Centre](#)

Widely Used Scientific Databases for Literature Search

Common sources researchers search before conducting an SLR

1



Google Scholar

2

Scopus[®]

Scopus

3



Web of Science[™]

4

IEEE Xplore[®]

IEEE Xplore

5

ScienceDirect[®]

ScienceDirect

6



PubMed

7



ACM Digital Library

8



SpringerLink

9

WILEY
Online Library

Wiley Online Library

10



MDPI



Common search sources vary by discipline.

Table 1
Summary of the state-of-the-art RNNs-based review papers.

Study	Gradient Descent-Based Optimization	Adaptive Learning Rate Methods	LSTMs-Specific Optimization Techniques	Weight Initialization Strategies	Metaheuristic Algorithms
Vanitha and Jayashree. (2023)	✓	X	✓	X	X
Narkhede et al. (2022)	X	X	✓	✓	✓
Fantin and Hadad (2022)	X	X	X	X	X
Alhumoud and Al Wazrah (2022)	X	X	X	X	X
Boulila et al. (2022)	X	X	X	✓	X
Lalapura et al. (2021)	✓	✓	X	X	X
Zeebaree et al. (2021)	X	X	✓	✓	X
Weerakody et al. (2021)	✓	X	✓	X	X
Abdulrahman et al. (2021)	X	X	✓	X	X
Torres et al. (2021)	✓	✓	✓	X	X
Gallicchio and Scardapane (2020)	X	X	X	✓	X
Van Houdt et al. (2020)	✓	X	✓	X	X
Sezer et al. (2020)	✓	✓	✓	X	X
Surakhi et al. (2020)	✓	✓	✓	X	X
Yu et al. (2019)	X	X	✓	X	X
Bianchi et al. (2017)	✓	✓	✓	✓	X
Salehinejad et al. (2017)	✓	✓	✓	✓	-
Greff et al. (2017)	✓	X	X	X	X
Lipton et al. (2015)	✓	✓	✓	X	X
Lukosevicius and Jaeger (2009)	✓	X	✓	✓	X
Prokhorov et al. (2002)	X	✓	X	X	X
This Study	✓	✓	✓	✓	✓

Paper:
[RNN-LSTM: From applications to modeling techniques and beyond—Systematic review - ScienceDirect](#)

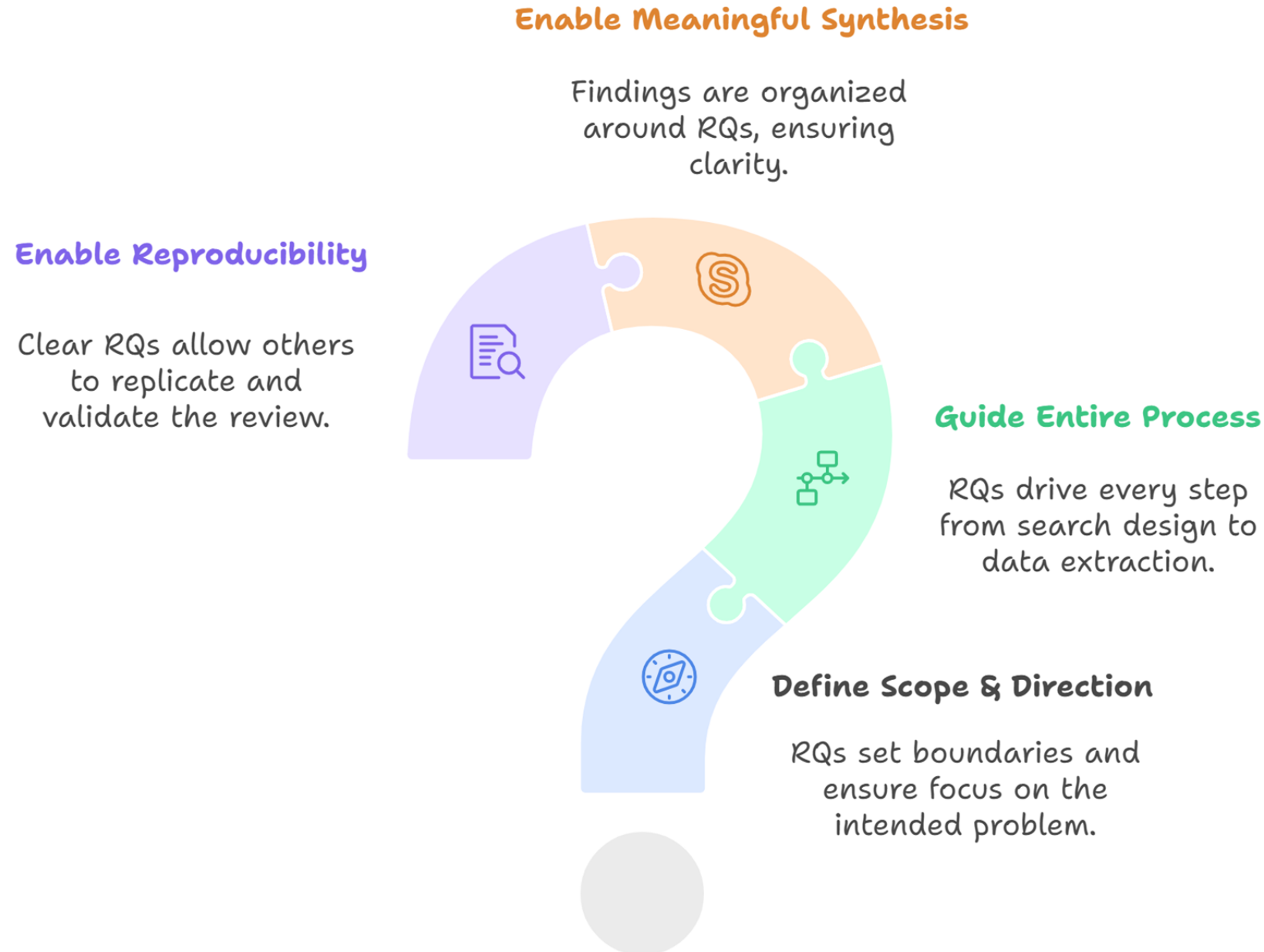
Table 1

Comparison of this work with existing healthcare explainability survey: Legends, X: Not discussed, ✓: discussed, ≈: partially discussed, DL: DL explanation techniques applied in medical health, ML: ML explanation techniques applied in medical health.

Ref.	Year	Focused applications	ML	DL	Datasets	Metrics	LIME in medical imaging	Challenges	Future direction
Holzinger et al. [69]	2017	Medical image and omic data segmentation	X	✓	X	X	X	X	≈
Adadi and Berrada [66]	2018	Current trends in explainable methods	✓	≈	X	X	≈	✓	≈
Mathews [68]	2019	Explainable AI uses in NLP, biomedicine, and malware categorization	✓	✓	X	X	✓	X	≈
Tjoa and Guan [64]	2020	Classification of XAI techniques and a brief discussion of their use in medicine	≈	≈	≈	X	≈	✓	✓
Singh et al. [24]	2020	Medical imaging for disease identification and prediction	X	✓	≈	X	X	≈	≈
Char et al. [67]	2020	Identifying moral issues for use in the field of healthcare	✓	≈	X	X	X	✓	✓
Adadi and Berrada [70]	2020	Transforming black boxes into interpretable models	✓	≈	X	X	≈	X	✓
Vilone and Longo [71]	2021	Concepts related to explainability and methods for assessing explainable AI	✓	≈	≈	✓	X	X	≈
Rasheed et al. [72]	2022	ML for healthcare that is comprehensible and reliable	✓	✓	≈	X	≈	✓	✓
Borys et al. [73]	2023	Saliency-based XAI method for clinical practitioners in medical imaging	X	✓	✓	X	X	X	✓
This study	2024	Model-Agnostic approaches in XAI for medical imaging analysis	✓	✓	✓	✓	✓	✓	✓

Paper:
[Local interpretable model-agnostic explanation approach for medical imaging analysis: A systematic literature review - ScienceDirect](#)

Step 1.2: Research questions formulation



Sample of
Research
questions
formulation

RQ1: What are the **current applications and domains** into which LSTM networks are used?

Motivation: To investigate the different usages of RNN-LSTM and how its learning ability impacts several domains and becomes state-of-the-art in numerous fields.

RQ2: What are the **current modeling techniques and tools** used to build the LSTM models?

Motivation: To explore the current modeling techniques and tools available for RNN-LSTM development including the programming languages, frameworks, libraries, and evaluation metrics.

RQ3: What are the **weights initialization techniques** used in LSTM networks?

Motivation: Weights initialization directly affects the network convergence. Thus, it is a crucial step before the training of LSTM models.

RQ4: What are the **weights optimization techniques** used in LSTM networks?

Motivation: To gain a comprehensive understanding of the most effective weight optimization techniques for RNN-LSTM networks.

RQ5: What is the **intensity of publications** related to LSTM?

Motivation: To investigate the frequency and volume of publications and the prevailing research trends concerning RNN-LSTM and suggest potential future directions.

This SLR examines the literature published over the past six years (2018–2023) and identifies the following as the most significant contributions:

- Presenting a systematic literature review with an **in-depth four-step approach** to methodically include studies concerning the initialization and optimization of LSTM weights (shown in Fig. 3).
- **RQ1:** Conducting an extensive discussion on the **current domains and applications into which LSTM is used**, including their architectures, datasets, evaluation metrics, strengths, limitations, and their contribution to the field (Section 5.1).
- **RQ2:** Identification of the **current dominated techniques and tools used to build the LSTM models**, including the preferred and most used programming languages (Section 5.2.1) and the most used evaluation metrics in regression and classification problems (Section 5.2.2).
- **RQ3:** Providing an **extensive analysis of various weight initialization techniques** specifically designed for RNN-LSTM networks (Section 5.3).
- **RQ4:** Comprehensively **summarizing numerous optimization algorithms used in optimizing LSTM networks** (Section 5.4).
- **RQ5:** Illustrating and **emphasizing key insights from the synthesized data** via a methodical mapping strategy to illustrate research intensity in the covered field (Section 5.5).

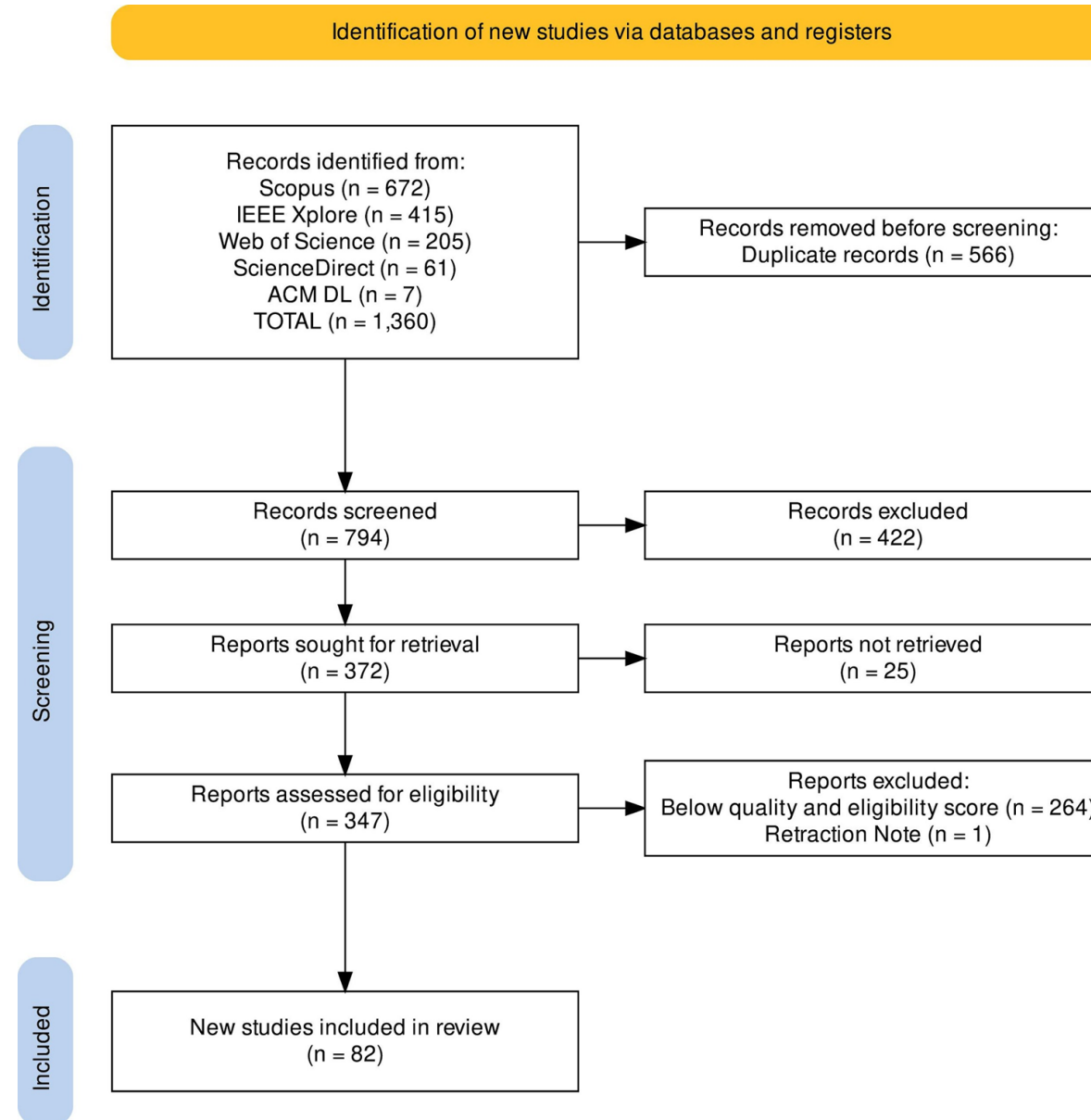
Step 1.3: Search criteria and strategies – The Boundaries

Databases Selected:	Scopus, IEEE Xplore, Web of Science, ScienceDirect, ACM Digital Library
Publication Types:	Journals, conference proceedings, book chapters
Timeframe:	2018–2023
Search Fields:	Title, abstract, and keywords
Keywords:	LSTM, “long short-term memory”, weight
Boolean Operators:	Used AND/OR and wildcards to refine search strings
Generalized Search String:	(*LSTM* OR “long short term memory”) AND weight
Date of Latest Search:	06/04/2023
Duplicates Removal:	Using EndNote 20.5 and Excel
Screening & Reporting:	Applied PRISMA standards, flowchart created via PRISMA tool

Step 1.3: Search criteria and strategies - Databases Search

Source	# of papers	URL	Search string
Scopus	672	https://bit.ly/lstm_slr_scopus	(TITLE(*LSTM* OR "long short term memory") AND TITLE-ABS-KEY(weight)) AND PUBYEAR > 2017 AND PUBYEAR < 2024 AND (LIMIT-TO (DOCTYPE,"ar") OR LIMIT-TO (DOCTYPE,"cp") OR LIMIT-TO (DOCTYPE,"ch")) AND (LIMIT-TO (LANGUAGE,"English")) AND (LIMIT-TO (SUBJAREA,"COMP") OR LIMIT-TO (SUBJAREA,"ENGI")) AND (LIMIT-TO (PUBSTAGE ,"final"))
IEEE Xplore	415	https://bit.ly/lstm_slr_ieeexplore	("Document Title":*LSTM* OR "Document Title":“long short term memory”) AND ("All Metadata":weight), Filters Applied (Type: Conferences or Journals, Years: 2018 – 2023)
Web of Science	205	https://bit.ly/lstm_slr_wos	*LSTM* OR "long short term memory" (Title) and weight (Topic), Refined by (Document Type: Article and Proceeding Paper, Language: English, Research Area: Computer Science or Engineering, Open Access)
Science Direct	61	https://bit.ly/lstm_slr_sciencedirect	Title: LSTM OR "long short term memory", Title, abstract, keywords: weight, Filters Applied (Type: Research articles and book chapters, Years: 2018 – 2023, Subject area: Computer Science and Engineering)
ACM Digital Library	7	https://bit.ly/lstm_slr_acmdl	[[Title: *LSTM*] OR [Title: "long short term memory"]] AND [Abstract: weight] AND [E-Publication Date: (01/01/2018 TO 12/31/2023)]

Step 1.3: Search criteria and strategies - PRISMA



PRISMA flowchart of the systematic literature process

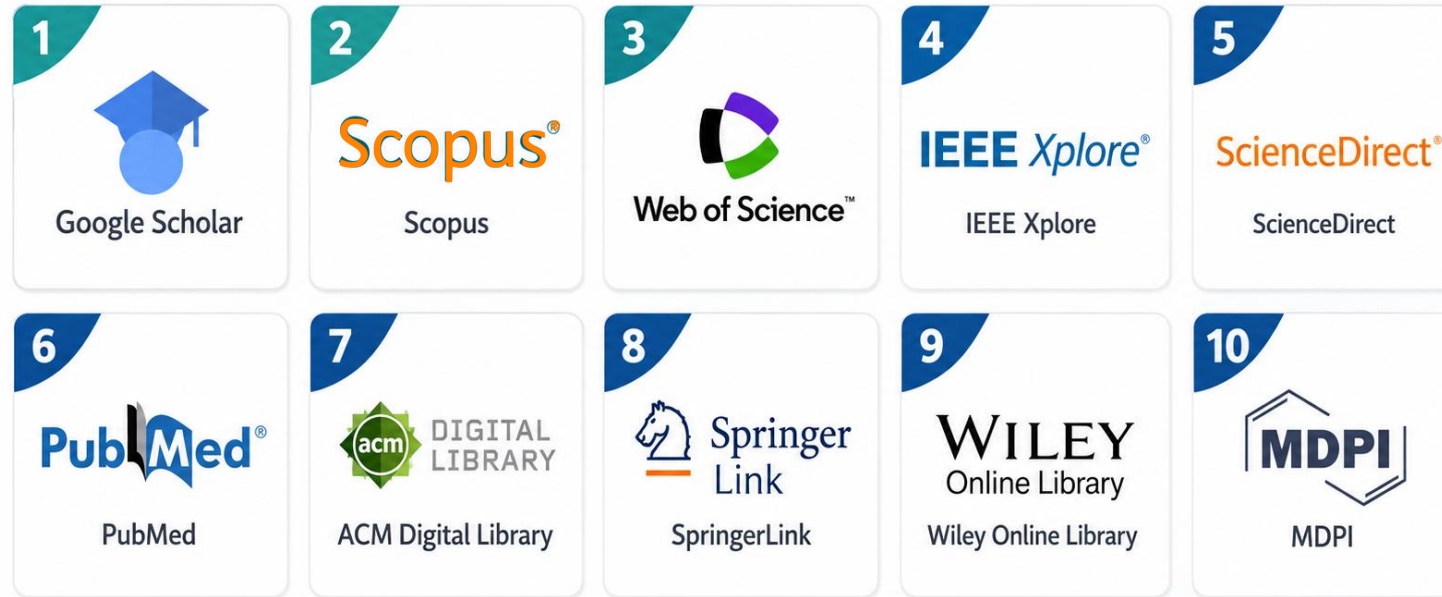
S.M. Al-Selwi et al.

Table 23
PRISMA 2020 checklist.

Topic	Item	Location
Title	1	Title
Abstract	2	Abstract
Introduction	3	Section 1
	4	Section 4.1.2
Methods	5	Section 4.3
	6	Section 4.1.3 , Table 5
	7	Sections 4.1.3 , 4.2.1
	8	Section 4.1.3
	9	Section 4.4
	10–15	Not applicable
Results	16a	Flowchart 5
	16b	Not applicable
	17	Section 5.1
	18–22	Not applicable
Other Information	24	Not applicable
	25	Acknowledgment
	26	Declaration of competing interest
	27	Data availability

Widely Used Scientific Databases for Literature Search

Common sources researchers search before conducting an SLR



Common search sources vary by discipline.

1 Search databases



Use keywords and filters to find relevant studies in the databases listed above.

2 Download results as CSV or RIS



Download either CSV or RIS files from the search results.

3 Import into reference manager

Import the downloaded files into reference manager tools to organize and manage references.

EN
EndNote™

Mendeley

Z
Zotero



Download either CSV or RIS files from the search results, then import them into reference manager tools.

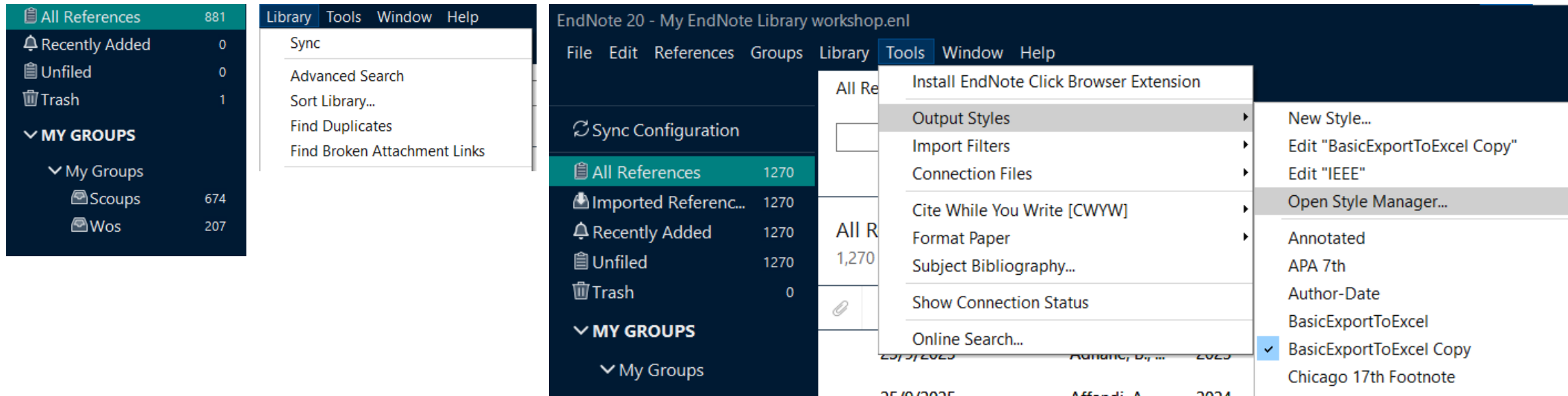


Tip: Exporting search results helps organize references, remove duplicates, and manage citations efficiently.

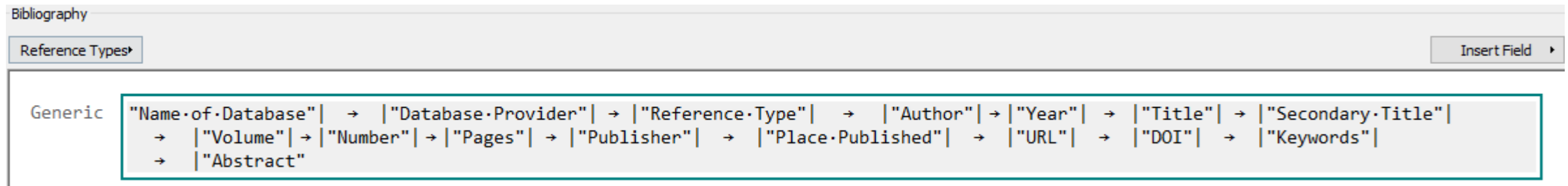


Step 1.3: Search criteria and strategies – Option 2 (Export to EndNote and Excel)

1. Export all your results to EndNote RIS file, create custom group for each import, then remove duplicates:



2. Add this output style using EndNote style manager (Tools, output styles, open style manager, new style):



Step 1.3: Search criteria and strategies - Export to EndNote and Excel

3. Select all your finalized non-duplicated results in EndNote using **CTL+A**, **CTRL+K**, then **CTL+V** in Excel. Put all your identified papers in the first Excel's sheet "1 Identification".

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Name of	Database	Reference	Author	Year	Secondary	Volume	Number	Pages	Publisher	Place	Keywords	Title	Abstract	DOI	URL
2	ACM DL		Conferenc	J. Shen; X.	2021	Proceeding			209–215	Associatio	Beijing,	Attention	Cough-		10.1145/35	https://doi
3	ACM DL		Conferenc	P. Rong-O;	2020	Proceeding			Article 20	Associatio	Bangkok,	PM2.5	The Impact		10.1145/34	https://doi
4	ACM DL		Conferenc	C. Huang;	2019	Proceeding			53–57	Associatio	Nanchang,	Chinese	Attention-		10.1145/33	https://doi
5	ACM DL		Conferenc	A. ElSaid;	2018	Proceeding			13–20	Associatio	Kyoto,	ant colony	Using ant		10.1145/32	https://doi
6	ACM DL		Conferenc	T. Dang; H.	2020	Proceeding			Article 38	Associatio	Bangkok,		DeepVix:		10.1145/34	https://doi
7	ACM DL		Journal	M. Bi; Q.	2021	ACM Trans.	20	5	Article 77			BLSTM	Bi-		10.1145/34	https://doi
8	ACM DL		Journal	R. P.	2022	Acm	21	4		Associatio		Computati	Q-Learning	Tree-LSTM	10.1145/34	https://ww
9	IEEE Xplore		Journal	X. Ma; H.	2021	IEEE	Mar-00			Institute of		Brain	A New	This	10.1109/TI	https://ww
10	IEEE Xplore		Conferenc	Y. Yang; W.	2022	2022 4th			1-Jul			Load	Research	On the	10.1109/CE	
11	IEEE Xplore		Conferenc	L. Yang; J.	2022	2022 23rd			1-Jun			Location	MicroMILT	Driven by	10.23919/A	
12	IEEE Xplore		Conferenc	G. Yang; H.	2022	2022			1-Aug			Visualizati	Named	As a type	10.1109/IJC	
13	IEEE Xplore		Conferenc	M. Xu; S.	2021	2020 25th			2089-2096			Convolutio	ACCLVOS:	Semi-	10.1109/IC	
14	IEEE Xplore		Conferenc	N. T. H.	2022	2022 11th			522-528			Measurem	Multiple	Improving	10.1109/IC	
15	IEEE Xplore		Conferenc	M.	2022	2022 Sixth			95-102			Adaptation	External	Online	10.1109/IR	
16	IEEE Xplore		Conferenc	N. Regev;	2021	2021			1-Aug			Query	Approxima	Despite	10.1109/IJC	
17	IEEE Xplore		Conferenc	V. T.	2022	2022 IEEE			1-Jun			Training	BILLNET: A	Long Short-	10.1109/Si	
18	IEEE Xplore		Journal	M. T. Khan;	2023	IEEE	70	1	266-279			Computer	Architectur	This paper	10.1109/TC	
19	IEEE Xplore		Conferenc	D.	2019	2019 Amity			228-234			Artificial	Compariso	The	10.1109/AI	
20	IEEE Xplore		Journal	M. Fazil; S.	2023	IEEE Access	11-Jan		16801-			Hate	Attentiona	Online	10.1109/AC	
21	IEEE Xplore		Conferenc	J. S.	2018	2018 15th			222-229			Videos	Context-	We	10.1109/CR	

< > ...
Reference Type
Setup
#Metrics
E&QA
Publisher
Techniques
4 Included
3 Eligibility
2 Screening
1 Identification
+

Step 1.3: Search criteria and strategies – Option 1 (Rayaan AI)

ra

Artificial Intelligence-Driven Pharmacogenomics for Precision DrugDos...

Need training? Upgrade

Blind mode

Screening criteria

OverviewReview dataScreeningFull text screeningData extraction

Showing 4 / 379 Undecided Articles

Undecided AND lstm

Filters

Undecided

Sort

1 Warfarin Dose Management Using Offlin...
Date: 2025-01-01
H. Ji; M. Gill; E. W. Draper; D. A. Liedl; D. ...

2 Long Short-Term Memory Network for D...
Date: 2022-01-01
Kuang, Y.; Liu, Y.; Pei, Q.; Ning, X.; Zou, Y.; L...

3 A Predictive Approach For Assessing M...
Date: 2025-01-01
Gupta, K.; Singh, J.; Singh, H.P.; Singh, M...

4 Long Short-Term Memory Network for D...
Date: 2022-01-01
Kuang, Y.; Liu, Y.; Pei, Q.; Ning, X.; Zou, Y.;...

Warfarin Dose Management Using Offline Deep Reinforcement Learning

Abstract:

Warfarin is a commonly prescribed anticoagulant with a narrow therapeutic window, which requires frequent and specialized monitoring. This work aims to develop standardized optimal warfarin dose decision support using a machine learning model based on time-series anticoagulation data and patient demographic characteristics. We propose an offline reinforcement learning model (RL) using a Batch-Constrained Q-Learning algorithm (BCQ) in a discrete action setting to predict the cumulative warfarin dose for the days until the next INR (International Normalized Ratio) test. Prior approaches utilized time-series supervised learning methods such as regression or Long Short Term Memory (LSTM) neural networks. The key advantage of reinforcement learning is its capacity to learn optimal dosing strategies from suboptimal clinical states in the data. To evaluate the model we compared the predicted warfarin doses with the physician-prescribed doses. Our BCQ model with a prediction accuracy of 98.6% significantly outperformed our baseline Long Short Term Memory (LSTM) model with a prediction accuracy of 71.09% . Further qualitative evaluation for explainability indicated that the model correctly adjusted the warfarin dose at time steps when patients had out-of-range INRs.

Publication Types:

Journal Article

Topics:

Reinforcement learning, Long short term memory, Medical treatment, Genetics, Predictive models, Q-learning, Biomedical monitoring, Data models, Data mining, Deep learning, Cardiovascular diseases, deep learning, LSTM, personalized medicine, Q-learning, reinforcement learning, thrombosis

Reason

Label

Analyze

Add note

Filters

controlled study 3

RCT 2

randomly allocated 2

double blinded 1

parallel group 1

randomised controlled trial 1

CCT 0

doubleblind 0

single blind 0

double marked 0

doubleblinded 0

single masked 0

control groups 0

parallel groups 0

controlled design 0

Keywords for exclude

Select All

cells 40

cohort 37

trials 33

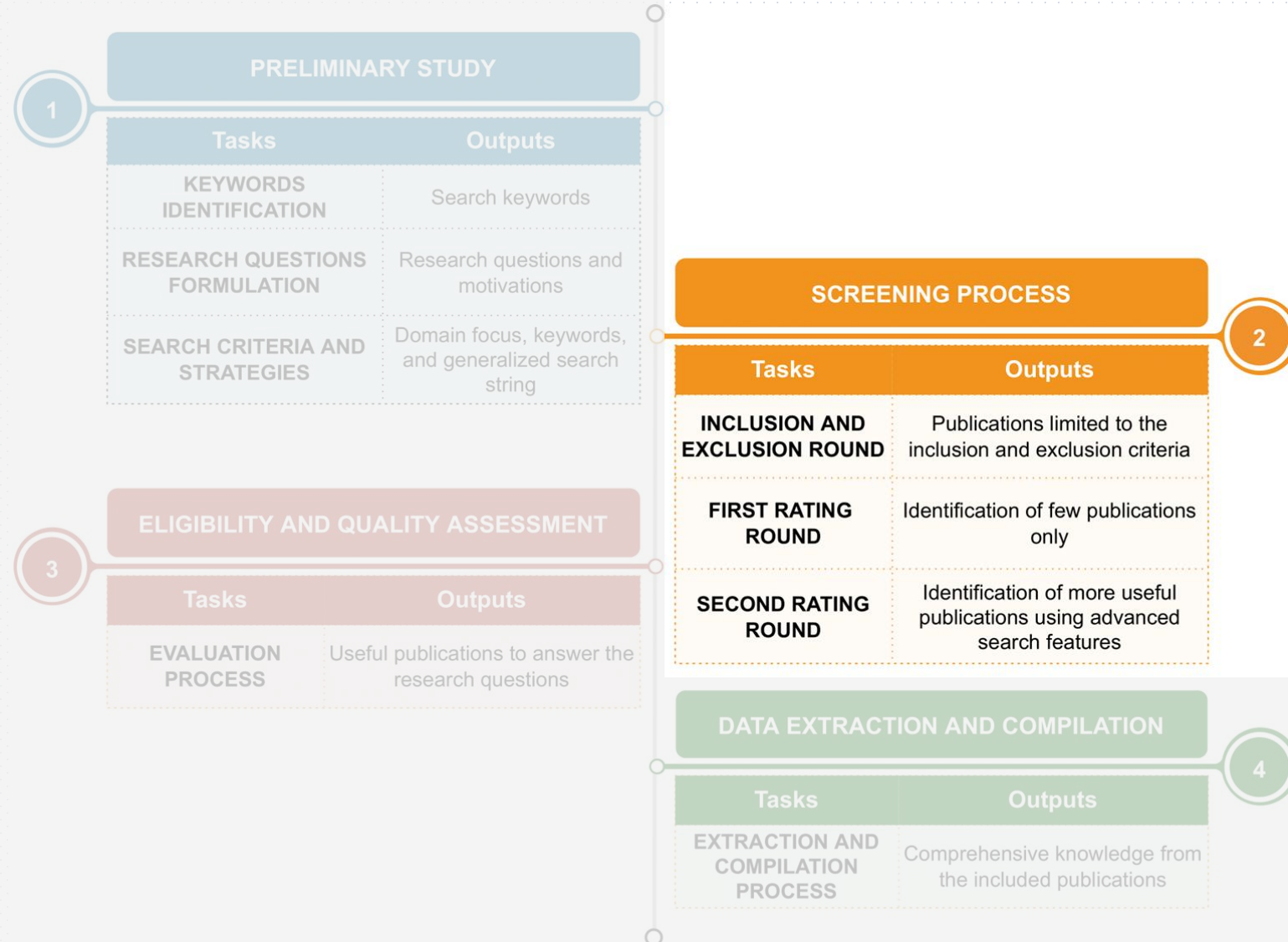
in vitro 19

Chat with ResearchPilot™

00hrs | 01mins | 01 Sessions

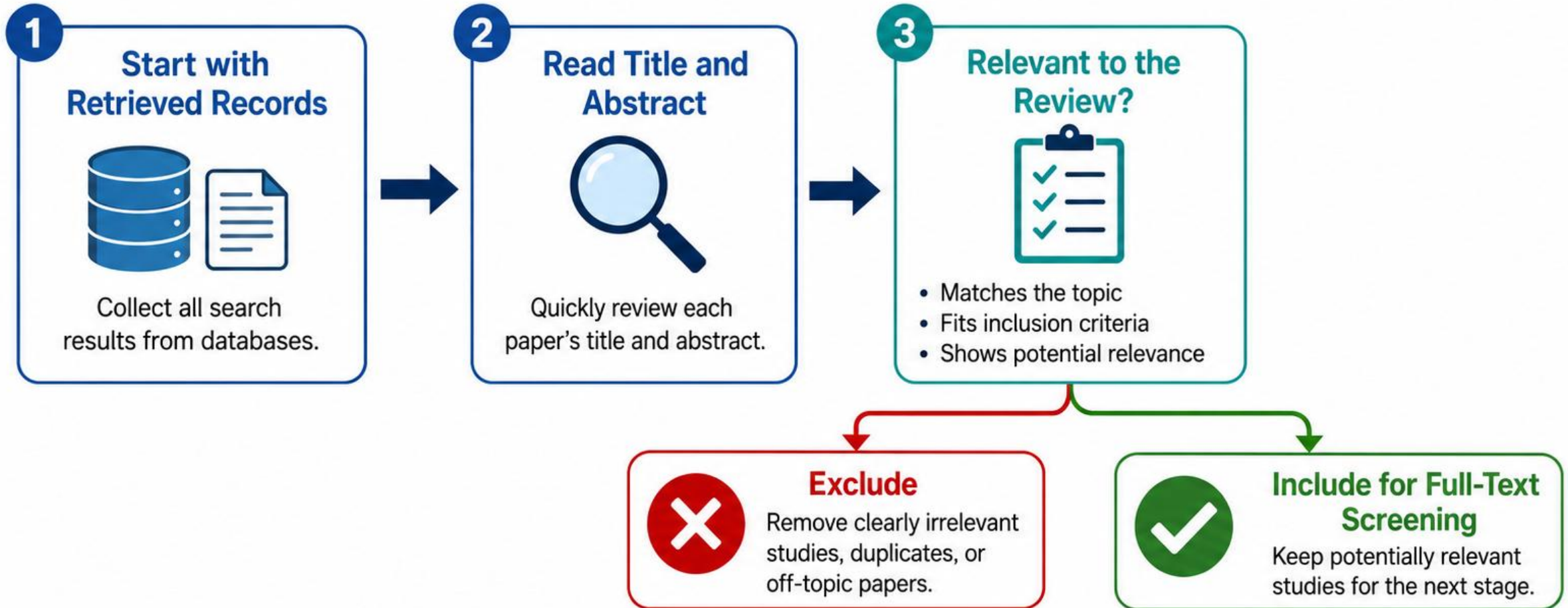
Step 2

Screening process



Title and Abstract Screening

A simple first filtering step in a systematic literature review



Purpose

To reduce a large number of records into a smaller set of potentially relevant studies.



Important

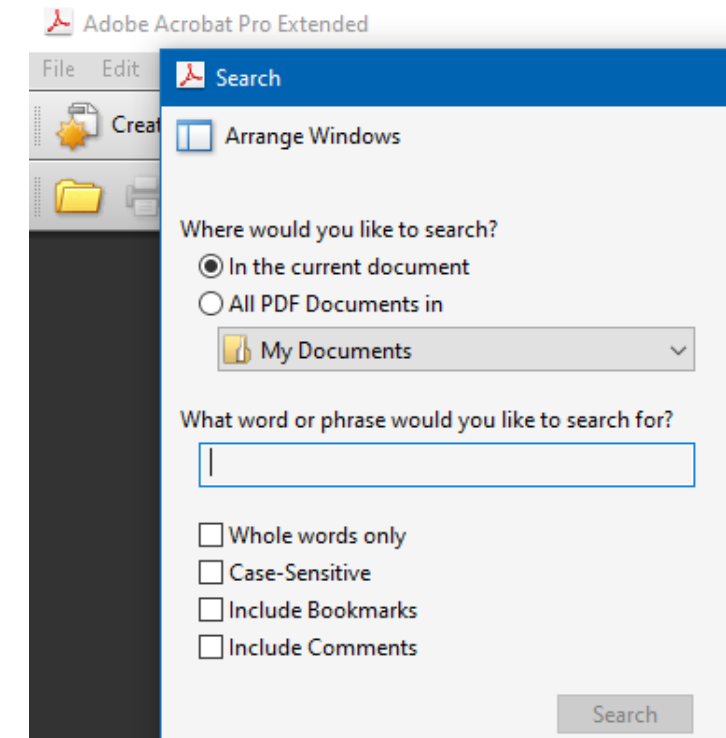
At this stage, decisions are based only on the title and abstract — not the full paper.

You can use advance search tools in Adobe Acrobat or EndNote to find some insights

Table 3

Inclusion and exclusion criteria for search screening.

Criteria	Inclusion Criteria	Exclusion Criteria
Publication year	2018–2023	Not between 2018–2023
Article Type	Journal papers, conference proceedings, and book chapters	Review papers, tutorials, seminars, interviews, letters, or blogs
Language	English	Non-English
Availability	Studies available in full text	Studies not available in full text (less than 5 pages)
Relevance	Relevant to the RQs	Not relevant to the RQs
Area	Studies in Computer Science and Engineering areas only	Publications not in Computer Science and Engineering areas
Publication Stage	Final (Published)	Not published yet (Articles in press)
Access Type	Open access	Not open access



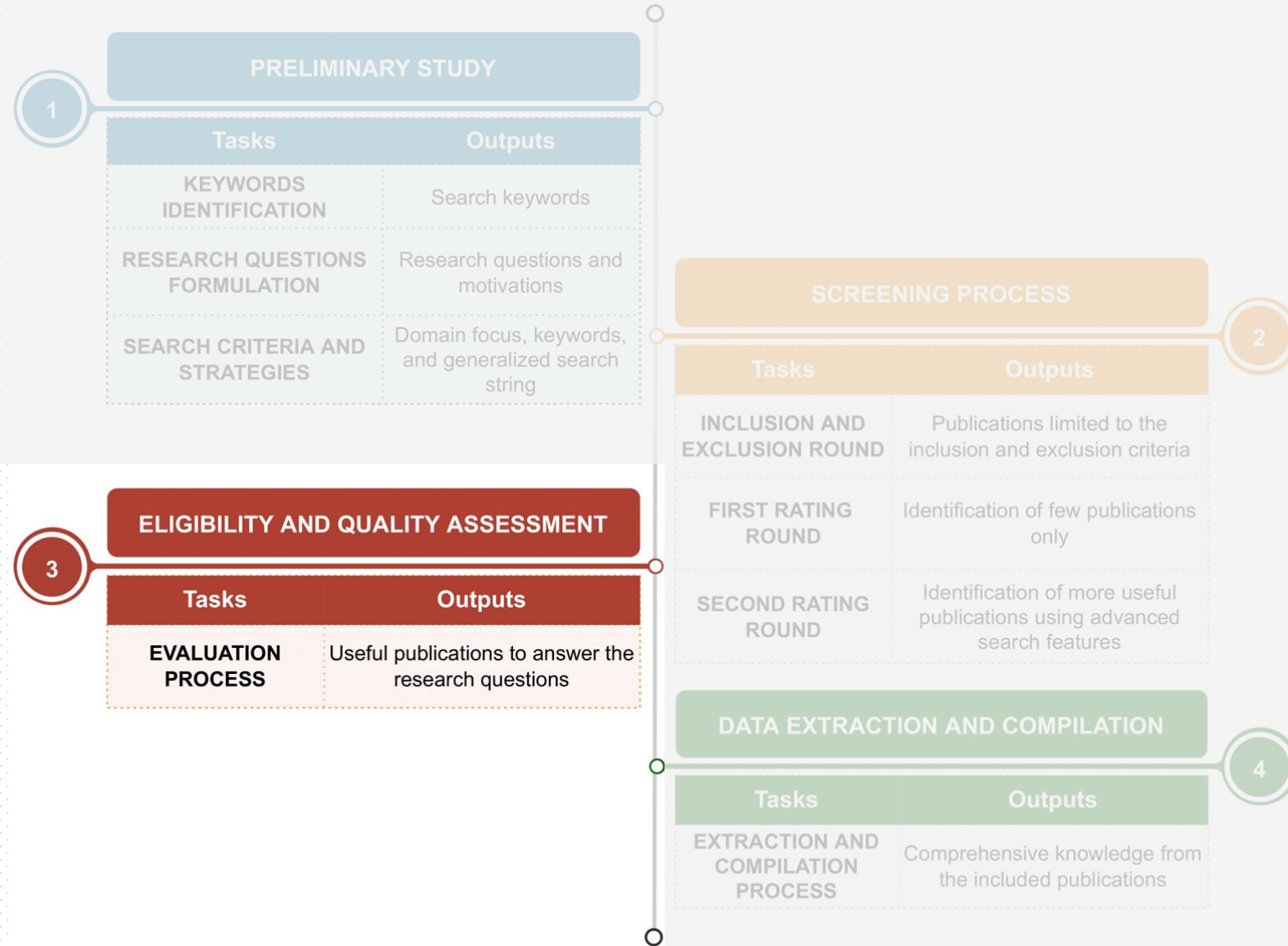
Step 2: Screening process

In Excel's sheet "2 Screening", start the screening process by reading title and abstract and then determine if paper will be further included or not.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W																				
	Name of Databases	Database Provide	Reference Type	Author	Year	Volume	Number	Pages	Publisher	Secondary Title	Place Published	Keywords	Title	Abstract	DOI	URL	Rating 1st Round	Rating 2nd Round	Final Rating	Note	Application	Methodology	Version																				
1	ACM DL		Journal	Zhang; M.	2021	20	5		n for	Transactio		term	Bi-	The	.org/10.114	w.scopus.c	☆	0	☆	1	☆	1	Question	BLSTM-																			
2	ACM DL		Conferenc	Zhang; W.	2021			209-215	n for	s Of The	China	mechanis	Cough-	Cough-	.org/10.114	.org/10.114	☆	0	☆	1	☆	1	Cough-																				
3	ACM DL		Conferenc	N.	2020			Article 20	n for	s Of The	Thailand	forecasting	The Impact	The last	.org/10.114	.org/10.114	☆	0	☆	1	☆	1	PM2.5																				
4	ACM DL		Conferenc	F. E. Jamiy;	2018			13-20	n for	-		optimizati	Using ant	This work	.org/10.114	w.scopus.c	☆	1		☆	1	prediction	ACO-LSTM	AC																			
5	ACM DL		Conferenc	Y. Chen; Q.	2019			53-57	n for	Internation		term	Attention-	Named	.org/10.114	w.scopus.c	☆	0	☆	0	☆	0	Chinese	AM-BiLSTM	AN																		
6	ACM DL		Conferenc	Van; H.	2020				n for	Internation		Science	DeepVix:	Machine	.org/10.114	w.scopus.c	☆	0	☆	0	☆	0																					
7	ACM DL		Journal	Hastuti; Y.	2022	21	4		n for	Transactio		onal	Q-Learning	Tree-LSTM	.org/10.114	w.scopus.c	☆	0	☆	0	☆	0																					
8	IEEE Xplore		Conferenc	K. M. Lim;	2021				Electrical	Internation		Convolutio	1D	Human	.org/10.110	w.scopus.c	☆	0	☆	1	☆	1	Human																				
9	IEEE Xplore		Journal	Xu; Q. Pan	2021	22	11	7242-7255		Transactio		Forestry	4-D Flight	The	.org/10.110	explore.ie	☆	0	☆	1	☆	1	Flight																				
10	IEEE Xplore		Journal	A. R. Zaidi;	2021	70		1-Sep		Transactio		extraction	A CNN-	Convolutio	.org/10.110	explore.ie	☆	0	☆	1	☆	1	Wrist																				
11	IEEE Xplore		Journal	Zhao; Y.	2022	10		88654	Electrical	IEEE Access		Forecastin	A	Multivariat	.org/10.110	w.scopus.c	☆	0	☆	1	☆	1	Multivariat	MLP-	AN																		
12	IEEE Xplore		Journal	Tawhid; S.	2022	71		1-Nov		Transactio		modeling	A	Epilepsy	.org/10.110	explore.ie	☆	0	☆	1	☆	1	Epilepsy																				
13	IEEE Xplore		Conferenc	Zhu; Y.	2020			1316-1321	Electrical	s - 2020		on (of	A Double	The CNN-	.org/10.110	w.scopus.c	☆	0	☆	1	☆	1	Text																				
14	IEEE Xplore		Conferenc	Zhang; Y.	2021			273-277	Electrical	Internation		g	A Dual-	As an	.org/10.110	w.scopus.c	☆	0	☆	1	☆	1	Remaining																				
15	IEEE Xplore		Journal	Yang; K.	2021	9		32822	Electrical	IEEE Access		gation	A feature-	Forecastin	.org/10.110	w.scopus.c	☆	0	☆	1	☆	1	tourists																				
16	IEEE Xplore		Journal	S. Kuenzel;	2022	71		1-Oct		Transactio		models	A Hybrid	In a	.org/10.110	explore.ie	☆	0	☆	1	☆	1	Combating																				
17	IEEE Xplore		Journal	F. Lin; X.	2021	22	11	6910-6920	Electrical	Transactio		g	A Hybrid	Accurate	.org/10.110	w.scopus.c	☆	0	☆	1	☆	1	short-	AM Conv-																			
18	IEEE Xplore		Journal		2022												☆	0	☆	1	☆	1																					
																							<	>	...	# Reference Type	# Setup	# Metrics	# E&QA	# Publisher	# Techniques	4 Included	3 Eligibility	2 Screening	1 Identification								

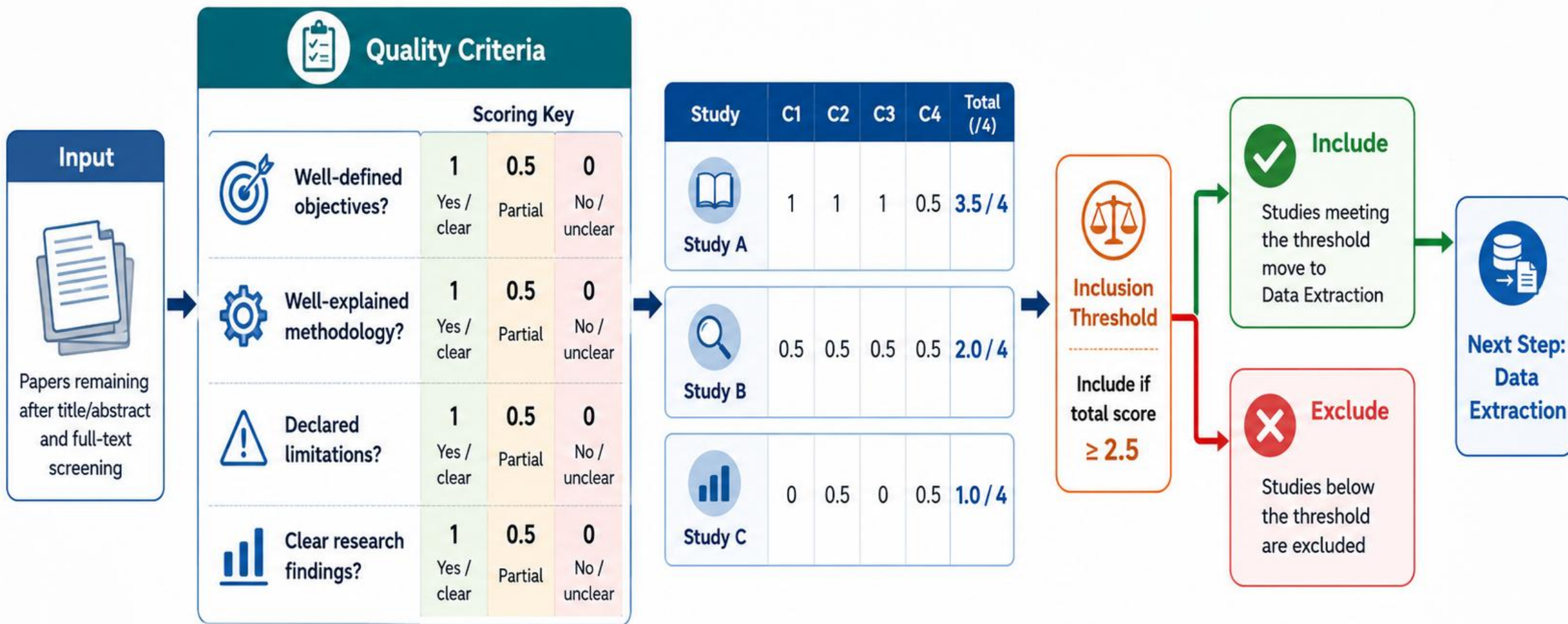
Step 3

Eligibility and quality assessment



Eligibility and Quality Assessment

Score each study against predefined quality criteria before final inclusion

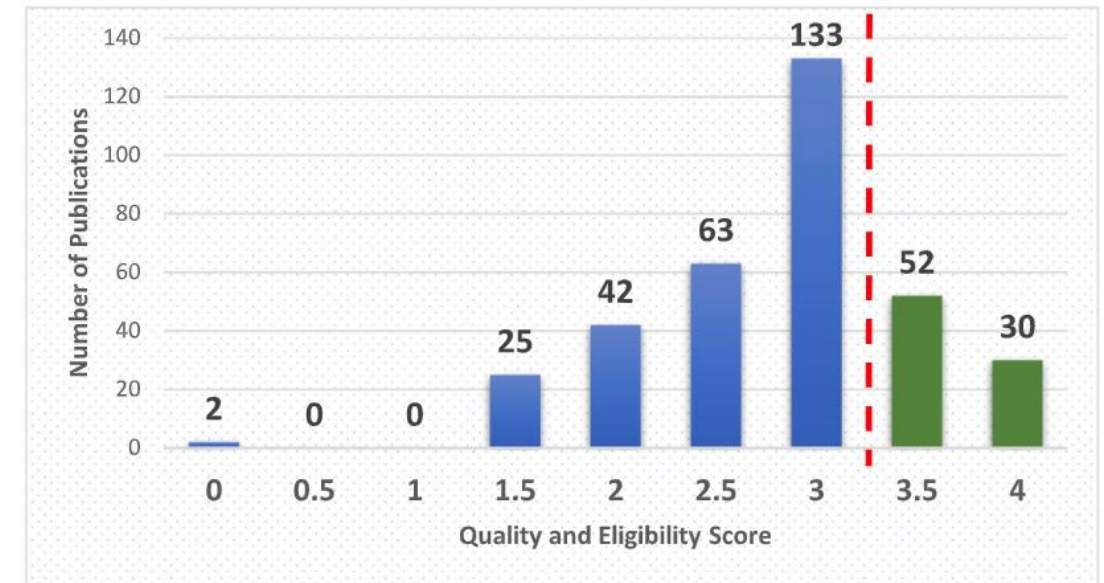


Quality metrics may vary by study, but each paper should be assessed using consistent criteria and a predefined threshold.

Step 3: Eligibility and quality assessment

Table 4. Criteria for scoring the eligibility and quality assessment.

Criteria	Score	Description
Well-defined objectives?	1	Indeed, the paper clearly outlines its objectives and goals.
	0.5	The paper outlines its objectives, though the goals are not explicitly defined.
	0	No, the paper does not have well-defined objectives.
Well-explained methodology?	1	Yes, the paper showcases a comprehensive, and well-documented methodology.
	0.5	Partially documented methodology that needs more details.
	0	No, the paper fails to provide a well-explained methodology.
Declared limitations?	1	Yes, the limitations are mentioned within the paper.
	0.5	The limitations are not in full detail.
	0	No, no declared limitations in this paper.
Clear research findings?	1	Indeed, the study provides clear findings with appropriate visualizations.
	0.5	The study does present its findings, but not in clear statements or visualizations.
	0	The study's research findings are unclear, and lacking elaboration.



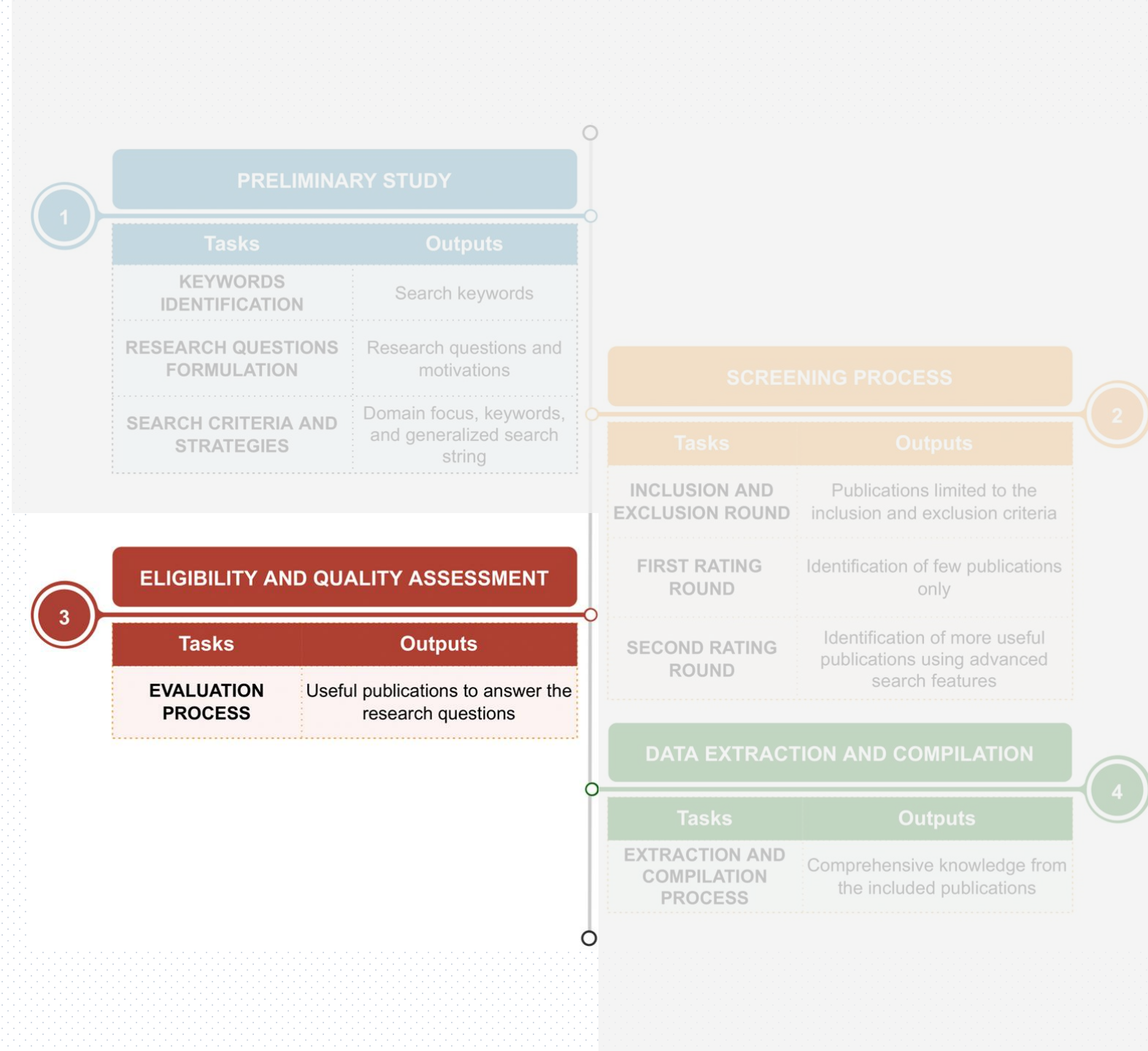
Step 3: Eligibility and quality assessment

In Excel's sheet "3 Eligibility", determine if paper will be further included or not based on your eligibility score.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W			
	Name of Databases	Database Provider	Reference Type	Author	Year	Volume	Number	Pages	Publisher	Secondary Title	Place Published	Keywords	Title	Abstract	DOI	URL	Application	Architecture	Objectives	Methodology	Limitations	Findings	Total			
1	ACM DL		Article	Zhang; M.	2021	20	5		n for	Transactio		term	Bi-	The intellig	https://doi.org/10.1145/3456789	https://www.acm.org/publications/papers/20210505	Question	BLSTM-SRLP	✓	1	✓	1	0.5	✓	1	3.5
2	ACM DL		e Paper	Zhang; W.	2021			209-215	n for	s Of The	China	mechanis	Cough-	Cough-base	https://doi.org/10.1145/3456790	https://doi.org/10.1145/3456790	Cough-		0.5	0.5	✗	0	0.5	1.5		
3	ACM DL		e Paper	N.	2020			Article 20	n for	s Of The	Thailand	forecasting	The Impact	The last few	https://doi.org/10.1145/3456791	https://doi.org/10.1145/3456791	PM2.5		✓	1	✓	1	✗	0	1	3
4	ACM DL		e	F. E. Jamiy;	2018			13-20	n for	-		optimizati	Using ant	This work e	https://doi.org/10.1145/3456792	https://www.acm.org/publications/papers/20180505	prediction	ACO-LSTM	✓	1	✓	1	✗	0	1	3
5	IEEE Xplore		e	K. M. Lim;	2021				Electrical	Internation		Convolutio	1D	Human activ	https://doi.org/10.1145/3456793	https://www.ieee.org/publications/papers/20210505	Human		✓	1	0.5	✗	0	1	2.5	
6	IEEE Xplore		Article	Xu; Q. Pan	2021	22	11	7242-7255		Transactio		Forestry	4-D Flight	The increas	https://doi.org/10.1145/3456794	https://ieeexplore.ieee.org/document/9456789	Flight		✓	1	✓	1	✓	1	4	
7	IEEE Xplore		Article	A. R. Zaidi;	2021	70		45170		Transactio		extraction	A CNN-	Convolutio	https://doi.org/10.1145/3456795	https://ieeexplore.ieee.org/document/9456790	Wrist		✓	1	✓	1	✓	1	4	
8	IEEE Xplore		Article	Zhao; Y.	2022	10		88654	Electrical	IEEE Access		Forecastin	A	Multivariate	https://doi.org/10.1145/3456796	https://www.ieee.org/publications/papers/20220505	Multivariate	MLP-	0.5	✓	1	0.5	✓	1	3	
9	IEEE Xplore		Article	Tawhid; S.	2022	71		45231		Transactio		modeling	A	Epilepsy (EP	https://doi.org/10.1145/3456797	https://ieeexplore.ieee.org/document/9456791	Epilepsy		0.5	✓	1	0.5	✓	1	3	
10	IEEE Xplore		e	Zhu; Y.	2020			1316-1321	Electrical	s - 2020		on (of	A Double	The CNN-LS	https://doi.org/10.1145/3456798	https://www.ieee.org/publications/papers/20200505	Text		0.5	0.5	0.5	0.5	0.5	2		
11	IEEE Xplore		e	Zhang; Y.	2021			273-277	Electrical	Internation		g	A Dual-	As an indis	https://doi.org/10.1145/3456799	https://www.ieee.org/publications/papers/20210505	Remaining		0.5	0.5	✗	0	1	2		
12	IEEE Xplore		Article	Yang; K.	2021	9		32822	Electrical	IEEE Access		gation	A feature-	Forecasting	https://doi.org/10.1145/3456800	https://www.ieee.org/publications/papers/20210505	tourists		✓	1	0.5	0.5	0.5	2.5		
13	IEEE Xplore		Article	S. Kuenzel;	2022	71		45200		Transactio		models	A Hybrid	In a conven	https://doi.org/10.1145/3456801	https://ieeexplore.ieee.org/document/9456792	Combating		✓	1	✓	1	✗	0	3	
14	IEEE Xplore		Article	F. Lin; X.	2021	22	11	6910-6920	Electrical	Transactio		g	A Hybrid	Accurate sh	https://doi.org/10.1145/3456802	https://www.ieee.org/publications/papers/20210505	short-Time	AM Conv-	0.5	✓	1	0.5	✓	1	3	
15	IEEE Xplore		Article	Kim	2022	10		34754	Electrical	IEEE Access		Commerce	A Hybrid	Accurate pri	https://doi.org/10.1145/3456803	https://www.ieee.org/publications/papers/20220505	Stock		✓	1	✓	1	✗	0	3	
16	IEEE Xplore		Article	S. F. Gau;	2020	14	2	299-311	Electrical	Journal On		dataset	A	The heterog	https://doi.org/10.1145/3456804	https://www.ieee.org/publications/papers/20200505	Classifying		✓	1	✓	1	0.5	✓	1	3.5
17	IEEE Xplore		Article	Y. Chen; W.	2020	8		11		Journal Of		bidirection	A Multi-	Background	https://doi.org/10.1145/3456805	https://www.ieee.org/publications/papers/20200505	Electrocardi		✓	1	0.5	✗	0	1	2.5	
18	IEEE Xplore		Article	Hu; Y.	2021	70			Electrical	Transactio		Forecastin	A New	This researc	https://doi.org/10.1145/3456806	https://www.ieee.org/publications/papers/20210505	Transformer	IDRSN-	✓	1	✓	1	✓	1	4	
19	IEEE Xplore		Article	Zhang; P.	2020	8		7295-7302	Electrical	IEEE Access		Iterative	A Novel	In this pape	https://doi.org/10.1145/3456807	https://www.ieee.org/publications/papers/20200505	CSI	CNN-AM-	✓	1	0.5	✗	0	0.5	2	
20	IEEE Xplore		Article	C. Liew; L.	2021	9		23671	Electrical	IEEE Access		Commerce	A Novel	Stock marke	https://doi.org/10.1145/3456808	https://www.ieee.org/publications/papers/20210505	Stock	IPSO-LSTM	✓	1	✓	1	✓	1	4	
21	IEEE Xplore		Article	J. Fan; T.	2020	8		155440	Electrical	IEEE Access		us vehicles	A Novel	Long short-t	https://doi.org/10.1145/3456809	https://www.ieee.org/publications/papers/20200505	Decision-	GOA-LSTM	✓	1	✓	1	✓	1	4	
22	IEEE Xplore		e	Zhu; M.	2019			100-105	Electrical	s - 2019 6Th		g	A	Short-term	https://doi.org/10.1145/3456810	https://www.ieee.org/publications/papers/20190505	Traffic flow	Regularized	0.5	0.5	✗	0	0.5	1.5		
23	IEEE Xplore		e	Chen; H.	2021			2854-2859	Electrical	Conferenc		forecasting	A Robust	The general	https://doi.org/10.1145/3456811	https://www.ieee.org/publications/papers/20210505	Wind Power		0.5	0.5	✗	0	0.5	1.5		
24	IEEE Xplore		e	Ruan	2021			249-253		Internation		learning	A	As a hot top	https://doi.org/10.1145/3456812	https://ieeexplore.ieee.org/document/9456793	sentiment		✓	1	0.5	✗	0	0.5	2	
25	IEEE Xplore		Article	Lin; Y.	2020	8		66066	Electrical	IEEE Access		Complex	A Sparse	The develop	https://doi.org/10.1145/3456813	https://www.ieee.org/publications/papers/20200505	Time Series	sparse	✓	1	0.5	0.5	0.5	1	2	
< > ... # Reference Type # Setup #Metrics # E&QA # Publisher # Techniques 4 Included 3 Eligibility 2 Screening 1 Identification + : ◀																										

Step 4

Data extraction



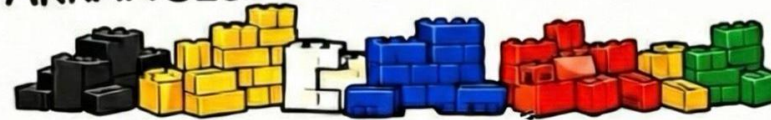
data



SORTED



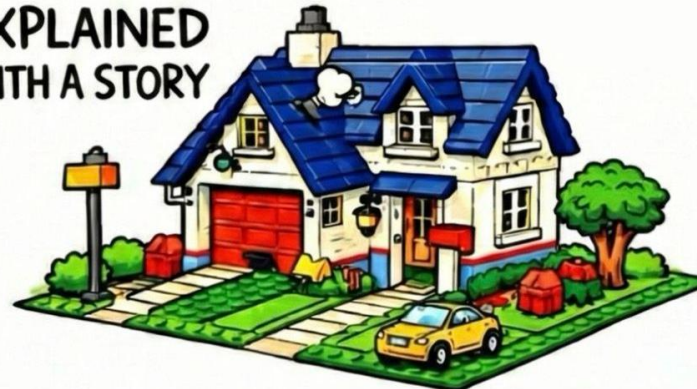
ARRANGED



**PRESENTED
VISUALLY**



**EXPLAINED
WITH A STORY**



Data extraction and compilation of included studies

In Excel's sheet "4 Included", extract the data you need from the finalized included papers.

RQ1: What are the current **applications** and domains into which LSTM networks are used?

RQ2: What are the current **modelling techniques and tools** used to build the LSTM models?

RQ3: What are the weights **initialization techniques** used in LSTM networks?

RQ4: What are the weights **optimization techniques** used in LSTM networks?

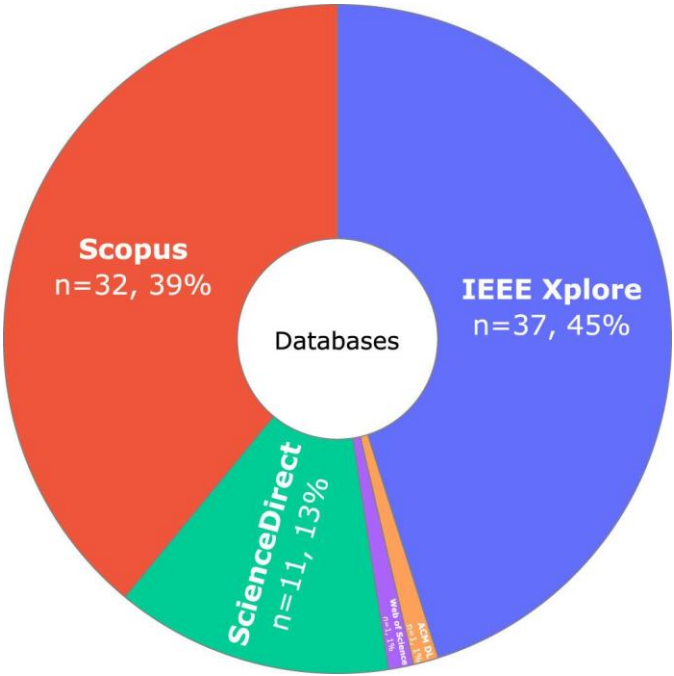
RQ5: What is the **intensity of publications** related to LSTM?

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC			
	Name of Databases	Main Publisher	Publisher	Secondary Title	Volume	Number	Pages	Place Published	Keywords	Reference Type	Authors	Year	Title	Abstract	DOI	URL	Weights 1st Round	Weights 2nd Round	Application	Domain	Architecture	Objectives	Motivation	Advantages	Limitations	Dataset	Evaluation Metric	Experiment Setup	Analysed	Verdict		
1	ACM DL	ACM	Association	Acm	20	5			Long	Journal	M. Bi; Q.	2021	Bi-	The	https://doi.org/10.1145/3456789	https://www.acm.org/publications/patents/10.1145/3456789		ADADELTA	Question	Natural	BLSTM-	To	Existing	Improved	Model	Food	[MRR],	Python	Ebrahim	Safwan		
2	IEEE	IEEE	Institute	2021 IEEE			808-812		Data	Confere	R. Ying;	2021	Predictio	How to	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		ADABOOS	Predictio	Stock	LSTM-	To	The	The LSTM-	The model	Historical	[RMSE],	-	Safwan			
3	IEEE	IEEE	Institute	2021					Brain	Confere	Z. C.	2021	Two-	Predictiv	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		SDAGD	Predictive	Prognosti	SDAGD-	To	The	The	Susceptib	Machiner	[MSE]	-	Safwan			
4	IEEE	IEEE	Institute	IEEE	9		23660-		Brain	Journal	Y. Ji; A.	2021	A Novel	Stock	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		IPSO	Stock	Stock	IPSO-	To	Predicting	Updating	It is not	S&P/ASX2	[RMSE],	Python	Safwan	Eb		
5	IEEE	IEEE	Institute	IEEE	8		155429-		Autono	Journal	Y. Shi; Y.	2020	A Novel	Long	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		GOA	Decision-	Computer	GOA-LSTM	To	To	Extract	The	NGSIM	Accuracy	Python,	Ebrahim	Saf		
6	IEEE	IEEE		IEEE	9		7701-		Task	Journal	A. Onan;	2021	A Term	Sarcasm	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		BO	Sarcasm	Natural	Three-	To	Sarcasm	The	The	Three	Accuracy	Java,	Ebrahim	Saf		
7	IEEE	IEEE	Institute	IEEE	9		47252-		Brain	Journal	J. Bian; L.	2021	Abnorm	In the	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		PSO	Abnormal	Energy	PSO-AM-	To	To	The	The model	The	[RMSE],	MATLAB	Ebrahim	Saf		
8	IEEE	IEEE		IEEE	9		30636-		Fire	Journal	Z. Y. Xu;	2021	Advance	Fire	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		ADAM	Fire	Safety	LSTM-VAE	To	To	Improve	The lack	69 real-	Fire	-	Ebrahim	Saf		
9	IEEE	IEEE		IEEE	11		16194-		Social	Journal	X. Wang;	2023	Adverse	Drug	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		AM	Adverse	Safety	Quantum	To	Leveragin	The QBi-	The need	TwoMed,	Precision,	Python,	Ebrahim	Saf		
10	IEEE	IEEE		IEEE	11		16801-		Hate	Journal	M. Fazil;	2023	Attentio	Online	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		AM	Hate	Speech	Multi-	To	Addressin	The	The model	Three	Accuracy,	Python	Ebrahim	Saf		
11	IEEE	IEEE		IEEE	9		5332-		Emotion	Journal	H. Y.	2021	Attentio	Speech	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		RMSprop	Speech	Speech	SCBAMM	To	Speech	SCBAMM	The study	EMO-DB	Accuracy,	Python,	Ebrahim	Saf		
12	IEEE	IEEE	Institute	IEEE	7		155130-		Bayesian	Journal	Y. Liu; G.	2019	Deep	In recent	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		SGD	Lithium-	Prognosti	BMA-	To	To	It	It is	CALCE	[ERUL]	[R]	Ebrahim	Saf		
13	IEEE	IEEE	Institute	IEEE	7		73627-		Brain	Journal	H. Wei;	2019	Named	Biomed	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		AM	ADAM	Biomedic	Natural	AM-	To	Previous	The	The	JNLPBA	Precision,	-	Safwan		
14	IEEE	IEEE	Institute	IEEE	8		165849-		Electric	Journal	L. Wang;	2020	Short-	Because	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		IPSO	Short-	Energy	OVMD-	To	To	The	It takes	The PV	[R ²],	Python	Safwan			
15	IEEE	IEEE		IEEE	7		107897-		Air	Journal	J. Ma; Y.	2019	Spatiot	As air	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		MBSGD	Pm2.5	Environm	IDW-	To predict	Deep	The	The study	Historical	[RMSE],	-	Safwan			
16	IEEE	IEEE	Institute	IEEE	9		59860-		Backpro	Journal	L. Yang;	2021	Speech	In recent	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		PSO	Speech	Speech	PSO-GA-	To	Tradition	Proposed	Truncatio	Two	Accuracy,	Python	Safwan			
17	IEEE	IEEE		IEEE	11		18245-		Credit	Journal	J. Eom; J.	2023	Time-	One of	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		ADAM	Predicting	Stock	Time-	To	By	The	The model	Three	Precision,	-	Safwan			
18	IEEE	IEEE	Institute	IEEE	9		146082-		Errors	Journal	X. Liu; Q.	2021	Using	Enhancin	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		AHMP	Predicting	Environm	AHMP	To predict	Tradition	The	The	396 data	[RMSE],	Python	Safwan			
19	IEEE	IEEE	Institute	IEEE	14	2	299-311		Large	Journal	Y. S. Lin;	2020	A	The	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		ADAMAX	Classifyin	Healthcar	Multimod	To	To	The	The	ADOS	unweighte	Python,	Ebrahim	Saf		
20	IEEE	IEEE	Institute	IEEE	22	6	6052-		Authenti	Journal	D.	2022	An ECG	The	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		AM	SGD	An ECG	Healthcar	Hierarchi	To design	To	Low	The	PTB ECG,	Accuracy,	Python,	Ebrahim	Saf	
21	IEEE	IEEE	Institute	IEEE	20	9	4868-		Indoor	Journal	Y. Zhang;	2020	An	In recent	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		SGD	Indoor	Communi	LSTM	To	The	The	The	Raw CSI	[MSE],	Python,	Ebrahim	Saf		
22	IEEE	IEEE	Institute	IEEE	20	23	14302-		Deep	Journal	J. Torres-	2020	Improv	Entities	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		ADADELTA	Detection	Safety	LeNet5-	To create	Detecting	The	the	A	Accuracy	-	Safwan			
23	IEEE	IEEE		IEEE	11	4	473-482		Anomaly	Journal	B. Yang;	2019	Anomalo	We	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		ADAGRAD	Anomalous	Computer	WCAE-	To	To	the	The	CUHK	[AUC],	✓ Not shared	Safwan			
24	IEEE	IEEE	IEEE	IEEE	69	3	441-452		Brain	Journal	X. Dai; H.	2020	Grow	Long	https://doi.org/10.1109/4281813.2021.9636038	https://ieeexplore.ieee.org/abstract/document/9636038		RANDOM	Image	Computer	Hidden-	To	LSTMs are	The	Requires	MSCOCO.	CIDER-D	PvTorch.	Safwan			
	<	>	...	# Reference Type	# Setup	# Metrics	# E&QA	# Publisher	# Techniques	4 Included	3 Eligibility	2 Screening	1 Identification	+													:					

Finally, it is time to condense the information extracted from the included studies into visual aids and provide answers to the previously posed RQs.

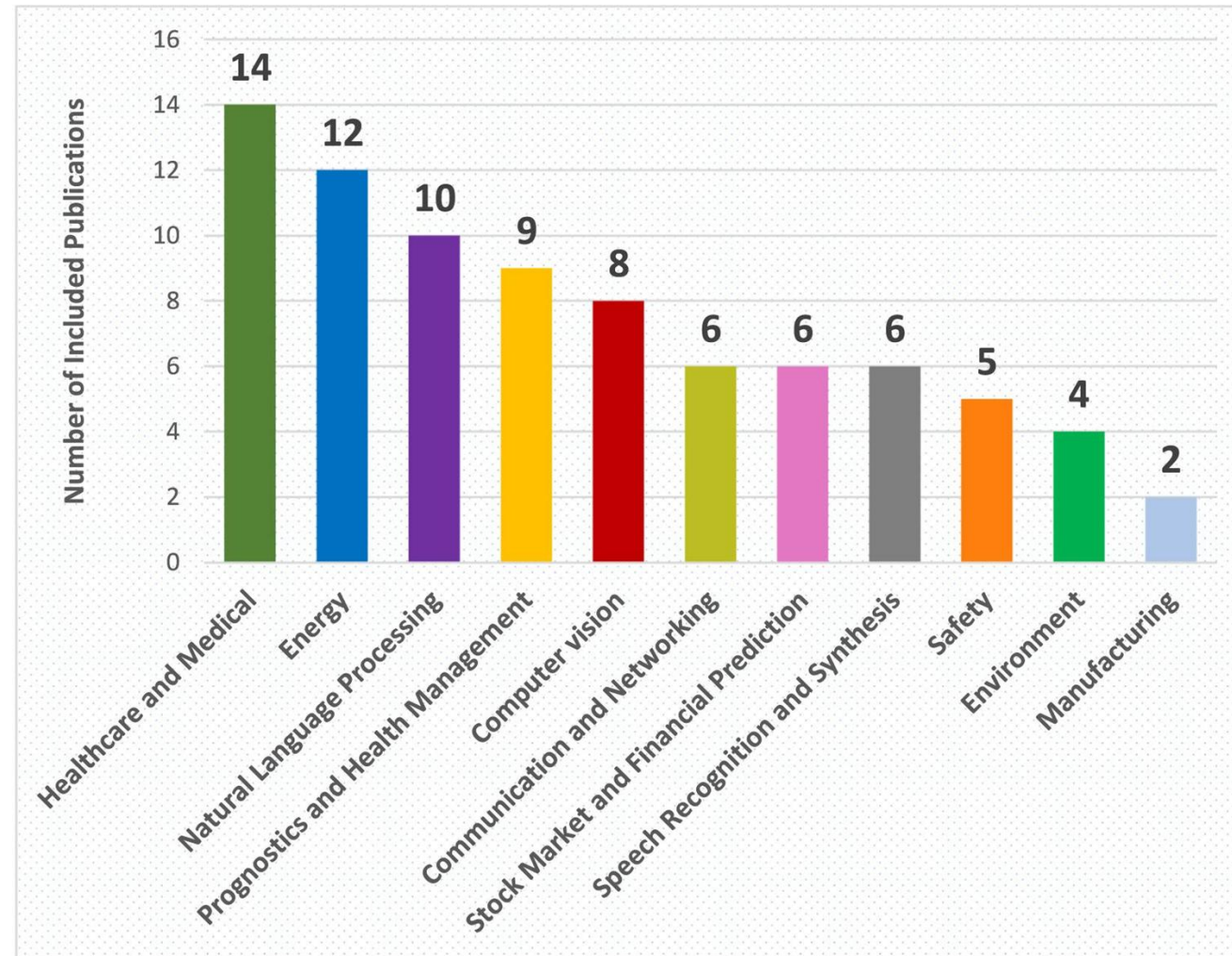
Table 5. Studies retrieved in each step of the systematic literature process.

Source	Identified	Unique	Screened	Pass E&QA	Included
IEEE Xplore	415	394	194	37	37
Scopus	672	307	100	32	32
Science Direct	61	57	40	11	11
Web of Science	205	29	9	1	1
ACM DL	7	7	4	1	1
Total	1360	794	347	82	82



Distribution of included studies by scientific databases and venues.

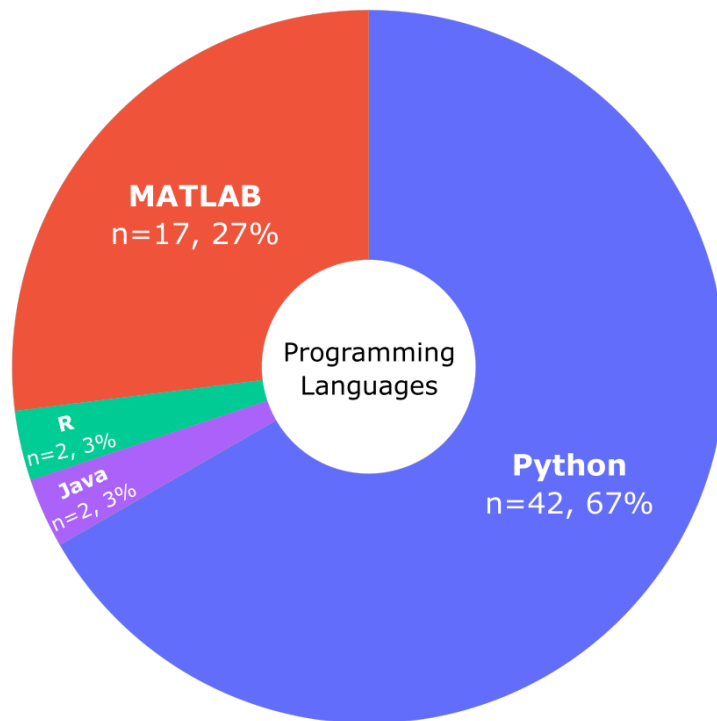
RQ1: What are the current applications and domains into which LSTM networks are used?



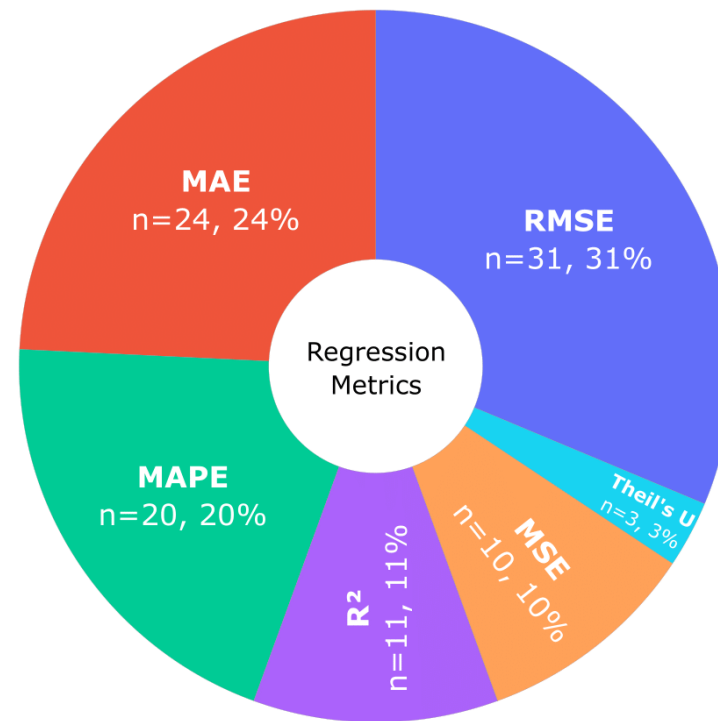
RQ2: What are the current modelling techniques and tools used to build the LSTM models?

Source: https://github.com/SafwanAlselwi/RNN-LSTM-SLR/blob/main/plotly_charts_saf1.ipynb

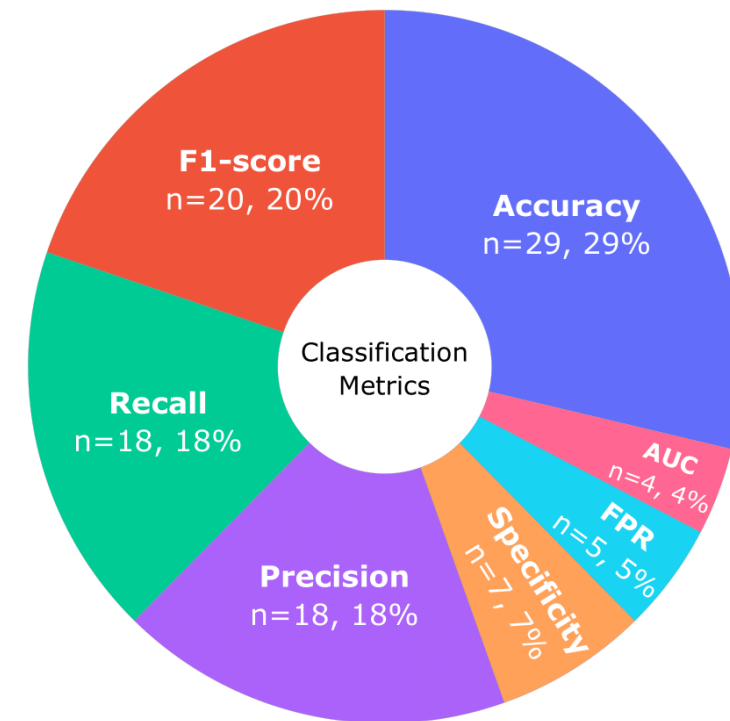
Generated using my Python's Plotly (Single Donut Chart)



Preferred programming languages for LSTM development

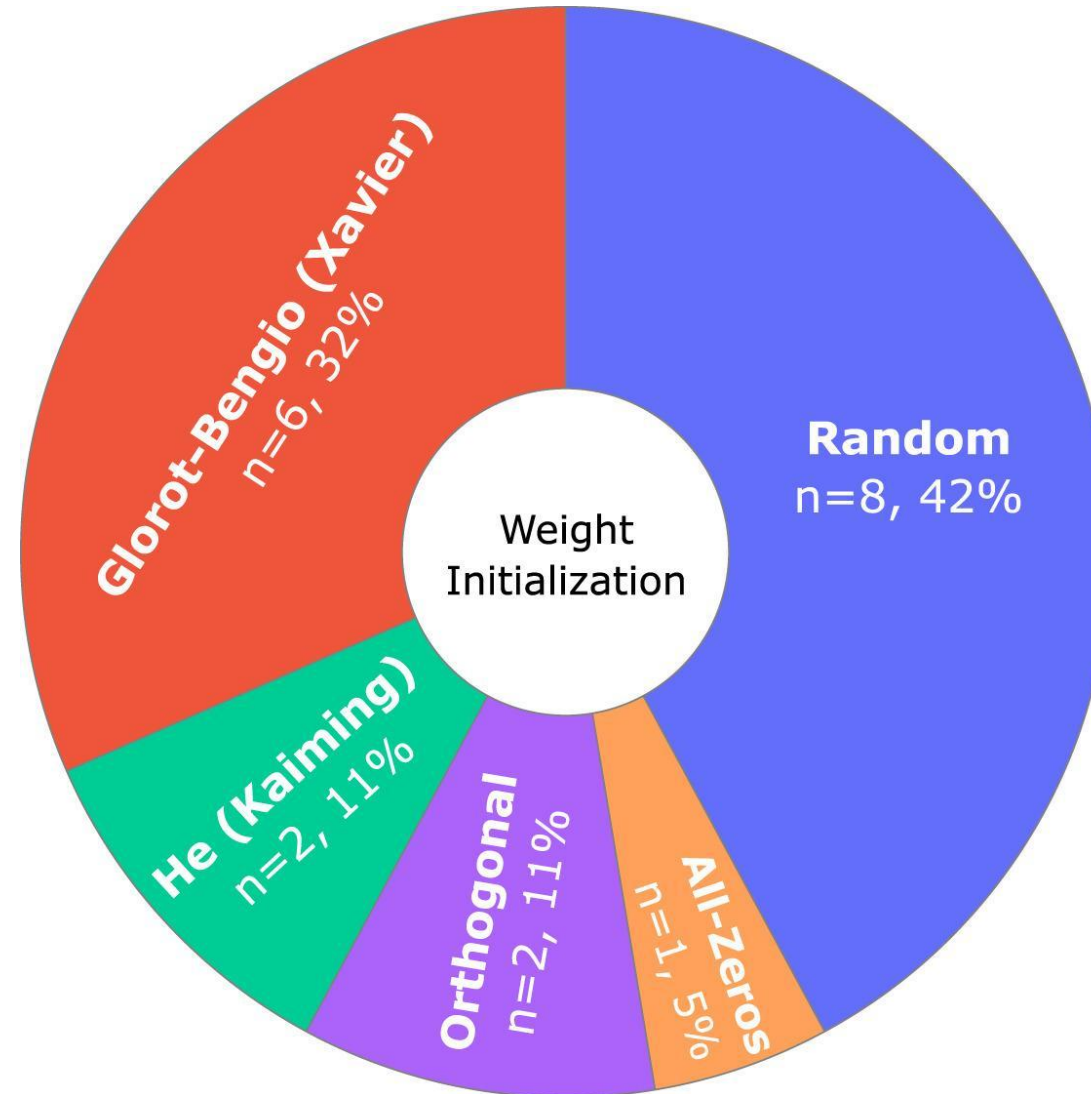


Most used evaluation metrics in regression problems

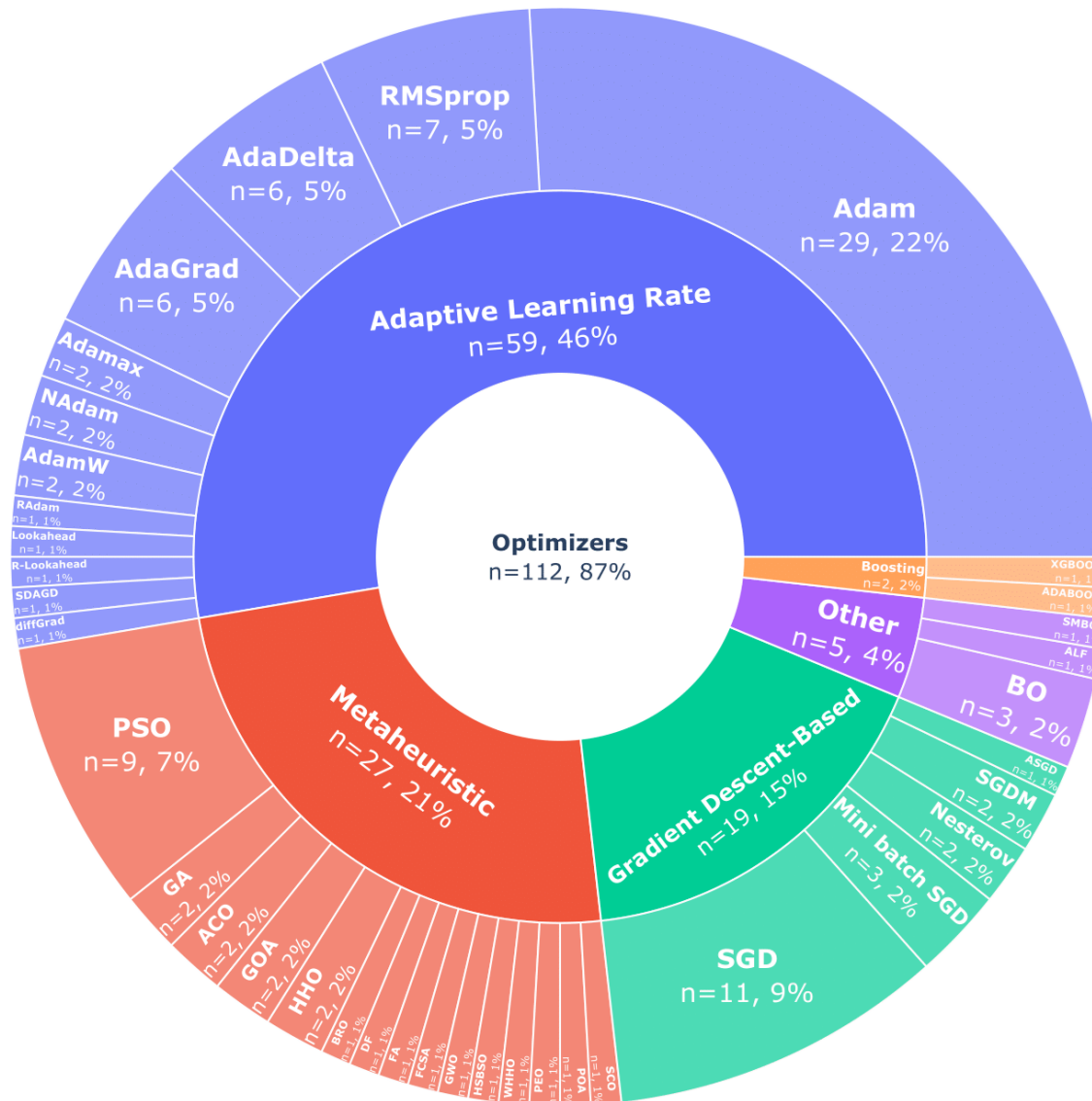


Most used evaluation metrics in classification problems

RQ3: What are the weights initialization techniques used in LSTM networks?

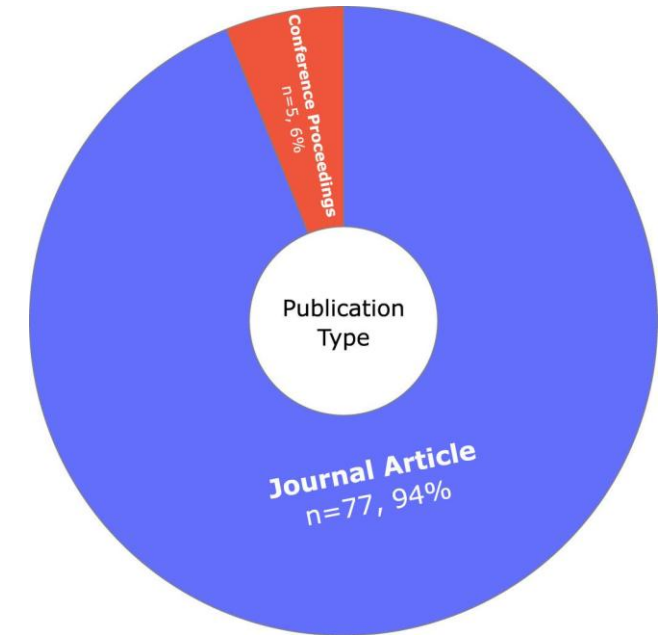
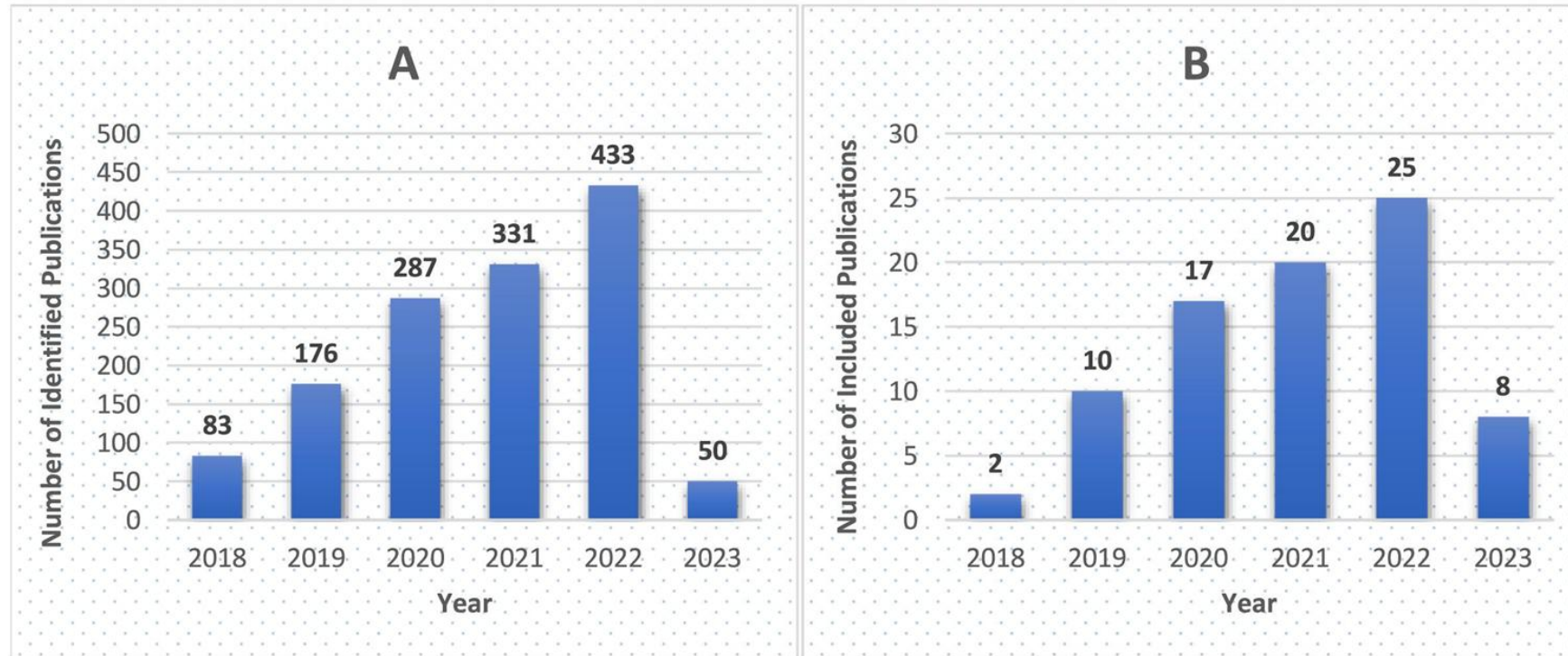


RQ4: What are the weights optimization techniques used in LSTM networks?



RQ5: What is the intensity of publications related to LSTM?

1. Distribution of included studies by **year**
2. Distribution of included studies by **publication type**.
3. Distribution of included studies by **publisher**.
4. Distribution of included studies by **journal**.



Extra

STRUCTURE OF RESEARCH PAPERS



VOSviewer - Visualizing scientific landscapes

What is VOSviewer?

- A software tool for constructing and visualizing **bibliometric networks** (e.g., co-authorship networks, keyword co-occurrence, citation networks) from your set of references.
- It helps reveal patterns like which keywords cluster together or which authors often collaborate, in your collected literature.

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- You can input your final set of included papers (from a RIS/BibTeX file) into VOSviewer.

Common analyses include

- **Keyword co-occurrence map:** shows clusters of terms that frequently appear together in titles/abstracts, indicating major themes.
- **Co-authorship map:** shows collaboration networks among authors or institutions in the field.
- **Citation or Co-citation map:** reveals how studies cite each other or share references, showing influential works.



The AI Dilemma: Support vs. Liability

Golden Rule: AI should support your thinking and writing – never replace it.

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- ✓ - Grammar/style suggestions
- ✓ - Summarizing long texts
- ✓ - Brainstorming research questions
- ✓ - Generating visuals (Napkin.ai)

Tools: ChatGPT, Grammarly AI, Scite.ai

Critical Limitations:

- ! - AI hallucinates fake citations
- ! - AI-written text lacks critical analysis
- ! - Journals require strict disclosure
- ! - Over-reliance kills skill development



Achievements

- Abdullahi, S., & et al. (2026). The rise of hallucination in large language models: systematic reviews, performance analysis and challenges. Cluster Computing, 29(2), 124.
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- Alghurabi, A., & et al. (2025). A systematic review of material erosion prediction techniques: Incorporating model parameters variability and the lack of field-scale representation. Results in Engineering, 107194.
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- Yalli, J. S., & et al. (2025). Authentication schemes for Internet of Things (IoT) networks: A systematic review and security assessment. Internet of Things, 30, 101469.
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- Sumiea, E. H., & et al. (2024). Deep deterministic policy gradient algorithm: A systematic review. Heliyon, 10(9).
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8

Article



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Al-Selwi, SM; Hassan, MF; (...); R
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